Axle Counters FLMF

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Note

The controlled documentation for Axle Counters is available via handheld SMS

Axle Counters FLMF
Section 1
General Principles

GENERAL PRINCIPLES

An axle counter is a device which counts the axles of a train into, and out of, a section of track. Its purpose is to detect the presence of a train. Note the important contrast with track circuits which prove the absence of a train. Axle counters are an alternative to track circuits, retaining the conventional signalling circuitry from the TPR onwards.

Axle counters have the following advantages over track circuits:

1. There is no need for in-section equipment such as cut section feeds and relays.

2. There are no cables connected to the rails, so there are no bonds and jumpers to be damaged.

3. Axle counters can provide train detection in places where ballast conditions are very poor (e.g. wet tunnels and coastal lines) and where track circuiting would be impossible or unreliable (e.g. iron-decked bridges).

4. A single axle counter section can be several miles long. This means that axle counters are very cost-effective on long stretches of plain line which would otherwise require several track circuits or cut sections.

Costs are reduced because equipment requirements are reduced (less cable and fewer lineside location cupboards are required) and maintenance and faulting times are also reduced since there is no need to inspect the whole section because there are no cables connected to the rails and no in-section equipment.

On completion of this section you will be able to:

1. Identify the components of axle counter systems.

2. Describe how an axle counter operates.

3. Use bonding plans to identify the limits of an axle counter section.

4. Distinguish between different types of axle counters.

In this section you will cover the operation and principles of the axle counter.

STRUCTURE

Axle counter systems consist of five main pieces of equipment.

1. A power supply unit, which can supply the equipment in the relay room and at the lineside

2. Count heads which detect the passage of the wheels

3. A junction box where the electronics associated with the count heads are located

4. The evaluator which is the master-mind of the system and performs the counting

5. Relays, which interface with the conventional signalling system.

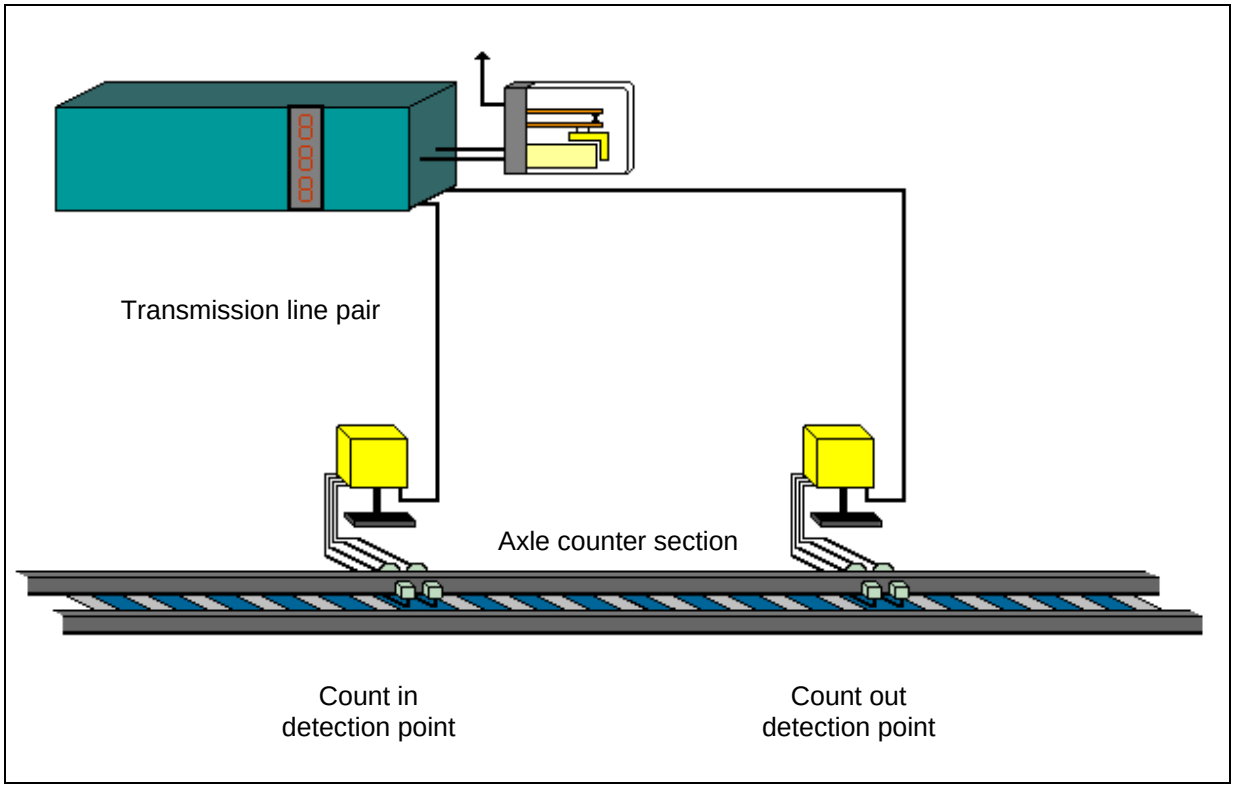


Figure 1.1 An axle counter section

Each end of an axle counter section is referred to as a detection point [ZP]. The rail-mounted equipment at a detection point consists of two count heads [SK]. These are mounted such that a wheel will always pass over SK1 before SK2 in the *NORMAL* direction of travel. Each count head comprises a transmitter coil (Tx) on the outside of the rail and a receiver coil (Rx) on the inside of the rail. The Tx coil is adjustable vertically on its mounting to enable the correct setting up of the count head to be carried out.

The lineside junction box is usually mounted to the left-hand side of the line. Equipment at all detection points is identical.

OPERATION

As a train enters the axle counter section, the wheels pass over the two count heads at the *count in* detection point. The first axle causes '1' to be stored in the counter at the evaluator, the equipment switches to the Track Occupied state, the feed to the TPR is cut off and the protecting signal is replaced to danger in the usual way. Each subsequent axle is counted and the total number of axles in the section is stored. Each axle leaving the section passes over the count out detection point and causes '1' to be subtracted from the number stored. The last axle clearing the section causes the stored number to return to zero, so the equipment returns to the Track Clear state, the TPR is re-energised and the signal clears.

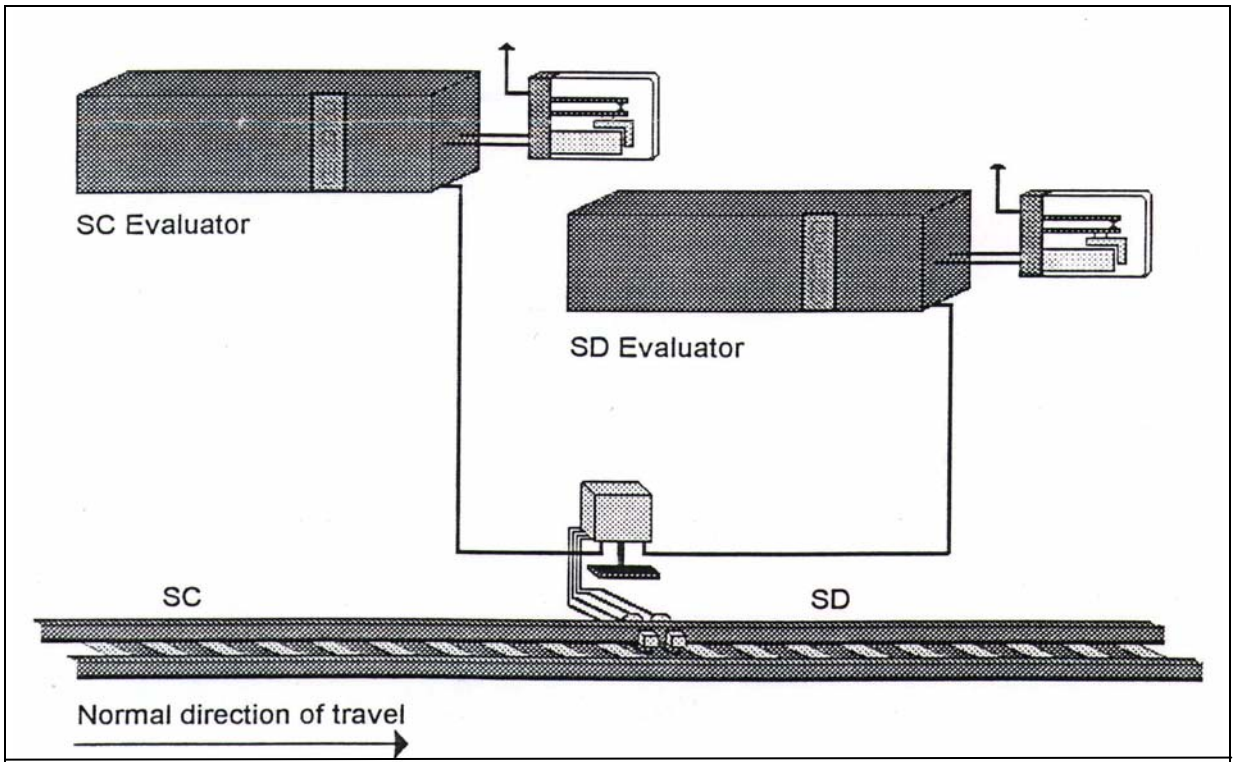
Being a replacement for track circuits, the axle counter must be able to respond to every type of train movement. Thus, it must show occupied if a train enters the section from either end (axle counters can therefore be used on bi-directional lines). It must also show occupied if a train enters and leaves the section at the same end (this means that axle counters can be used on a single-line branch or a bay platform).

As a train enters the axle counter section against the normal direction of travel, that is at the normal 'count out' end, the wheels pass over the count out count heads in the order count head 2 [SK2] then count head 1 [SK1]. The system detects that axles are passing the count heads in reverse order and thus entering not leaving the section. Therefore '1' is added to the counter store for each axle. At the normal 'count in' end, the train passes out of the section. Again the axles pass over the count heads in the order SK2 then SK1 so the system knows that the train is leaving the section and each axle causes '1' to be subtracted from the counter store.

Axle counters cannot cope with the "model train set scenario". If a trolley is lifted onto the track within an axle counter section, it will not be detected. The section will, however, show occupied when the trolley leaves the section. However, this sort of activity is permitted only during engineers' possessions. Note that we do not expect trolleys and other lightweight vehicles to operate track circuits at all.

Axle counter systems can have up to three sets of detection point equipment per evaluator. Most axle counter systems are (on Railways in Britain at least) on plain line and therefore have one detection point at each end. Where two axle counter sections abut, the detection point common to both will work both evaluators. In the diagram below, the detection point will count out for section SC using SC evaluator, but at the same time will count in for section SD using SD evaluator. The count store on SC evaluator decreases whilst the count store on SD evaluator increases.

Figure 1.2 Two axle counter sections abutting



PLAN SYMBOLS

Axle counters are usually shown on signalling and bonding plans by the symbols below, but you may come across local variations. The same symbols are used for both single and double rail configurations. The arrow denotes the Normal (reference) direction.

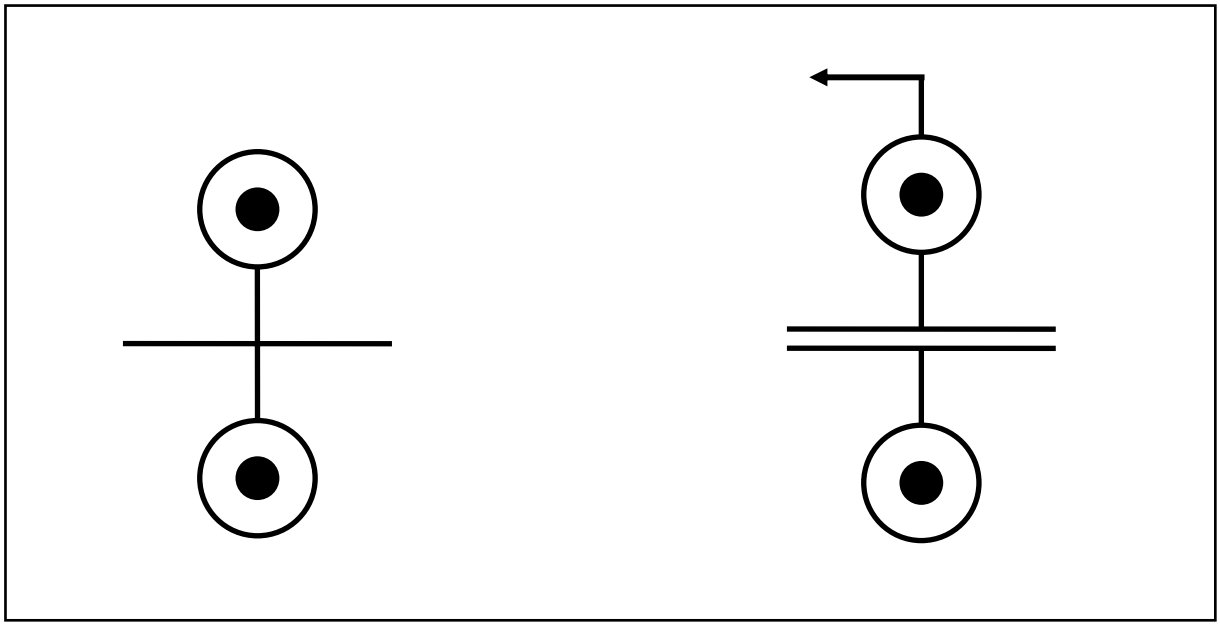


Figure 1.3 Signalling and bonding plan symbols

To the signalman an axle counter section appears on the panel as a normal track circuit, the only difference being the suffix (A.C) after the identification letters.

DOUBLE AND SINGLE RAIL SYSTEMS

Position of Count Heads

There are two configurations of axle counter, double rail and single rail. The Az 65 version and the original AzL 70 version are both double rail. In this case the count heads are positioned two on a single rail a fixed distance apart. This distance (which is fixed) is chosen by the manufacturer to ensure that the system is able to detect both the smallest acceptable wheel at the highest speed.

Two count heads are used to detect direction of traffic and number of axles that have passed over the count heads. Consider the count in end of the section. If an axle passes the count heads in the order ‘SK 1’ then ‘SK 2’, the system registers that the axle has moved into the section. If the axle passes the count heads in the order ‘SK 2’ then ‘SK 1’, the system registers that the axle has passed out of the section.

With single rail systems, there is one count head mounted on each running rail which are staggered from each other but not doubled up.

The Four Types

The single rail systems are all AzL 70 variations. To distinguish the AzL 70 versions apart, they are sometimes given the letters A, B, C and D by railways in Britain.

1. Type A is the double rail version.

2. Type B is a single rail version, which uses the Type A evaluator with amended configuration strapping and a later type of junction box called EAK 30. All single rail versions use this type of junction box, hence they are referred to as AzL 70/30 versions.

3. Type C is also single rail. It has an electronic junction box similar to the Type B, but some of the cards in the evaluator are different. This type is often referred to as the high frequency or audio frequency type.

4. Type D is similar to the Type C, but the outputs from the junction box to the evaluator are at lower frequencies so some of the cards in the junction box and evaluator are slightly different. The lower transmission frequency allows the maximum achievable length of the axle counter section to be increased. However, this version does have a lower maximum operational speed. The junction box is sometimes known as the EAK 30S, and this type is often referred to as the voice frequency type.

Because of slight variations as updating has occurred, it is important that you check the part codes on any cards before replacing them. Although the cards might look the same and have the same name, if the codes are different you must not interchange them.
Further information can be found in Appendix I.

Advantages of Double Rail Types

Having said all that, the count heads in all the AzL 70 versions are similar. Since they comprise separate transmitter and receiver units which are bolted together through the web of the rail, the only physical difference between the double and single rail AzL 70 count heads is the different layout on the track.

Single rail systems have several important advantages over double rail systems. These are shown below.

Single rail	Double rail
Cables between the count head pairs	No cables crossing to the other rail
Stagger distance has to be measured by the installer	The separate units abut each other, so the distance is pre-set
Rail creep can be a problem	Rail creep is not a problem with the count heads on one rail

Table 1.1

This module covers the AzL 70/30 high frequency, or Type C. References are made to the other versions where appropriate. The single rail systems are very similar from the maintenance and faulting points of view and your instructor will be able to point out the differences where they exist.

Axle Counters FLMF
Section 2
The Outdoor Equipment

THE OUTDOOR EQUIPMENT

This section looks at the structure and operation of the count heads and the electronic junction box.

On completion of this section you will be able to:

1. Describe the function of the count heads and junction box.

2. Describe how the count heads and junction box are linked into the system.

COUNT HEADS

The Equipment

The two count heads [SK] at a detection point are bolted to the cess rail 148mm apart. They are mounted as one unit split about the centre hole so this dimension is fixed. The whole unit is insulated from the rail.

The transmitter coil is bolted to the outside of the rail, but you can adjust the position of the transmitter coil during maintenance and fault-finding to enable the best possible output from the receiver coil. The transmitter moves up and down in a series of small steps. Note that it must never touch the rail. Poor transmitter alignment results in less effective wheel detection particularly for the faster trains and trains with smaller wheels.

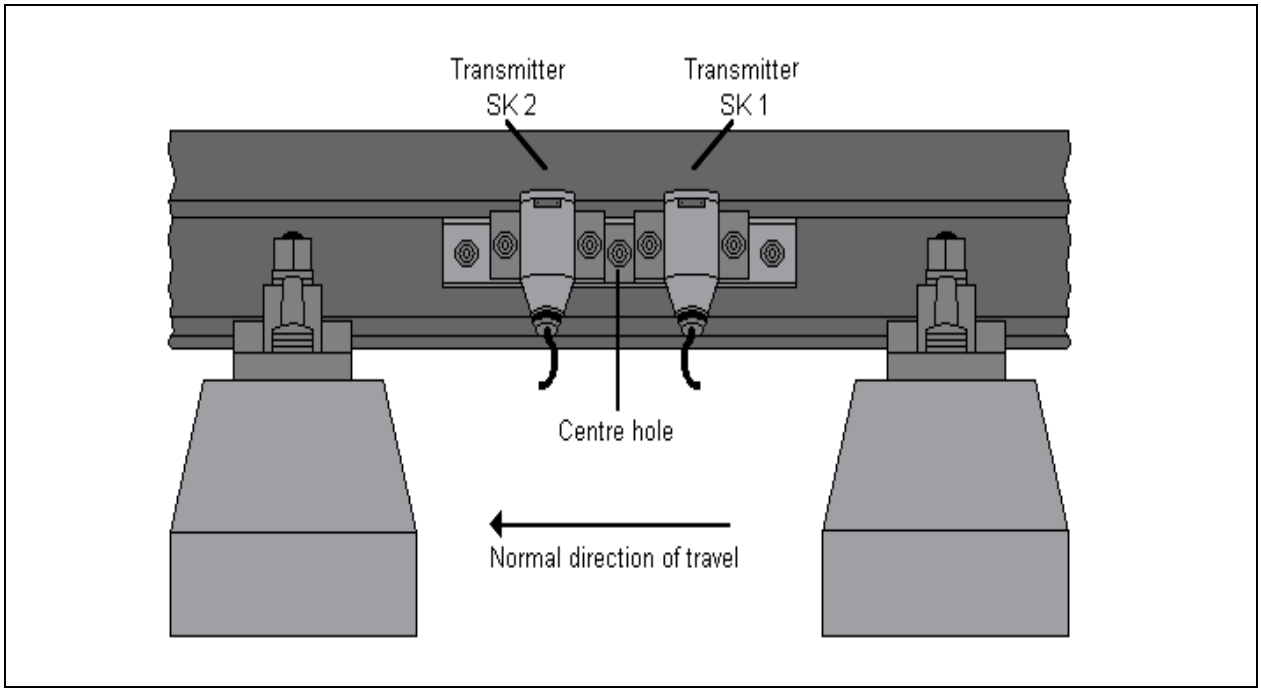


Figure 2.1 The count heads

There are two count heads at each detection point so that each passing wheel is detected twice. The evaluator in the relay room or REB senses the direction in which the wheel is travelling over the detection point by the sequence in which the two count heads are influenced. Each detection point can therefore cause the evaluator to count in or count out, depending on the direction in which the train is travelling. This ensures that the system can accommodate a train passing through the section in either direction or entering and leaving the section at the same end. Axle counters can, therefore, also be used on single line branches and bay platforms.

The cables which connect the count head to the electronic junction box are 4 metres long and must never be reduced in length because their a.c. electrical characteristics are taken into account in the design of the transmitter and receiver oscillator circuits. Also, if the cable is damaged or cut, the moisture seal will be broken.

If a detection point is between two adjacent axle counter sections and therefore acting as both a count out for one section and a count in for the next section, the junction box is connected to the two evaluators, the one to rear counting down and the one in advance counting up as each wheel passes.

How a Rail Count Head Works

Each count head consists of two coils, a transmitter coil (Tx) on the outside of the rail and a receiver coil (Rx) on the inside of the rail. The transmitter coil is fed with a.c. from an oscillator in the junction box and produces an alternating magnetic field. This field envelopes the rail and the receiver coil. The strongest lines of magnetic flux reaching the receiver, 'cut' its coil at angle a in such a way as to induce an a.c. voltage in it. This voltage is in phase with the transmitter voltage.

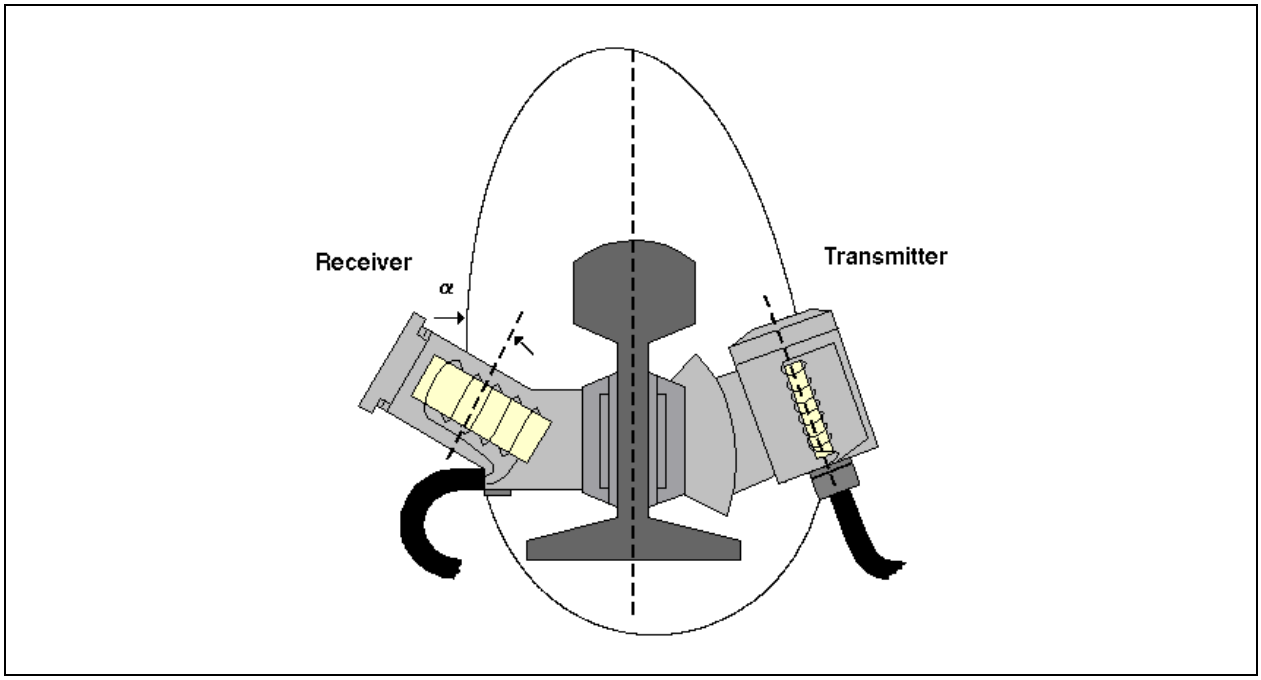


Figure 2.2 Count head showing main flux line with no wheel present

Therefore the receiver produces maximum ‘in-phase’ voltage. If it helps, you could think of a count head as working in a similar way to a transformer, the core being air instead of iron. The ‘in phase’ message is passed back to the junction box where it is interpreted as ‘No wheel over count head’.

As a wheel approaches the count head, this bulk of metal begins to distort the magnetic flux pattern and the receiver output voltage decreases. When the wheel is directly over the count head, it distorts the magnetic field so much that the strongest available flux at the receiver coil ‘cuts’ the coil in the opposite direction compared to when there was no wheel present. As a result the receiver output voltage is now 180° out of phase with the transmitter voltage. This message is interpreted by the junction box as ‘Wheel present over count head’.

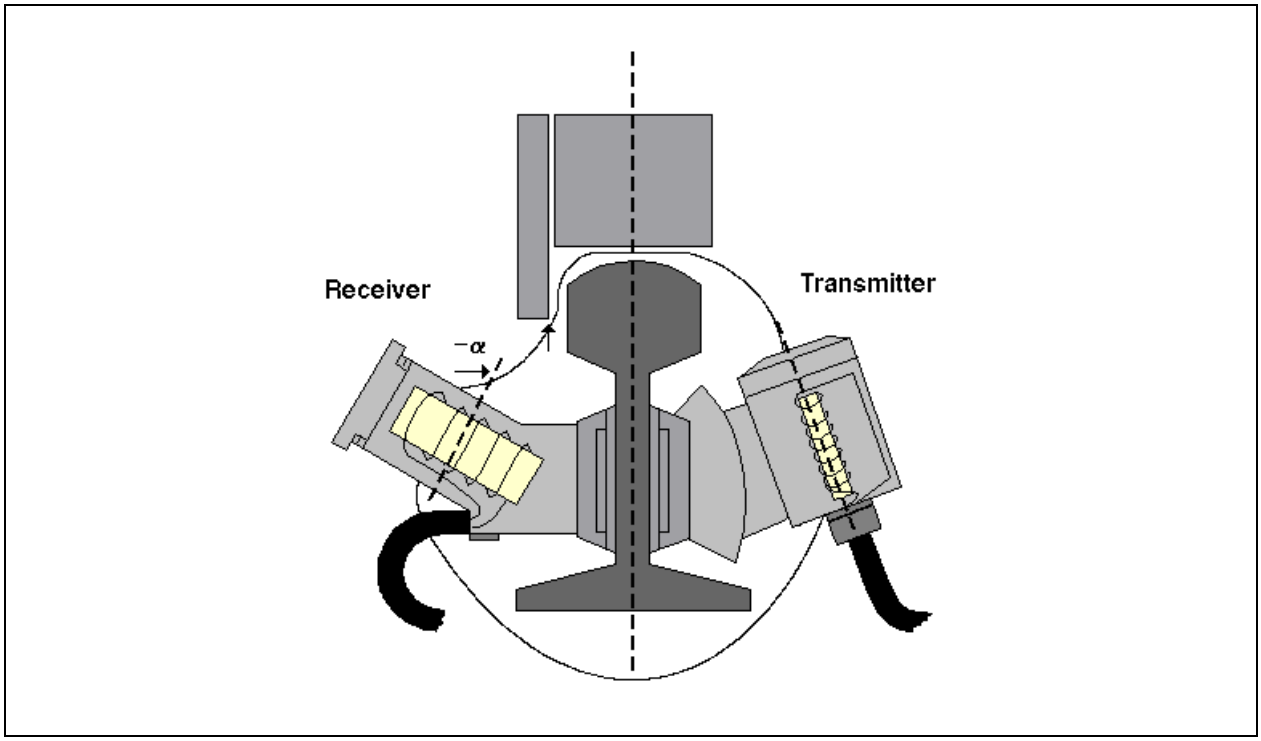


Figure 2.3 Count head with wheel directly overhead

By virtue of the explanation given, there must be a stage as the wheel approaches, when the strongest available flux at the receiver coil ‘cuts’ the coil at 90°. At this point the receiver output voltage is zero. This phase reversal is used by the electronic junction box to send information to the evaluator.

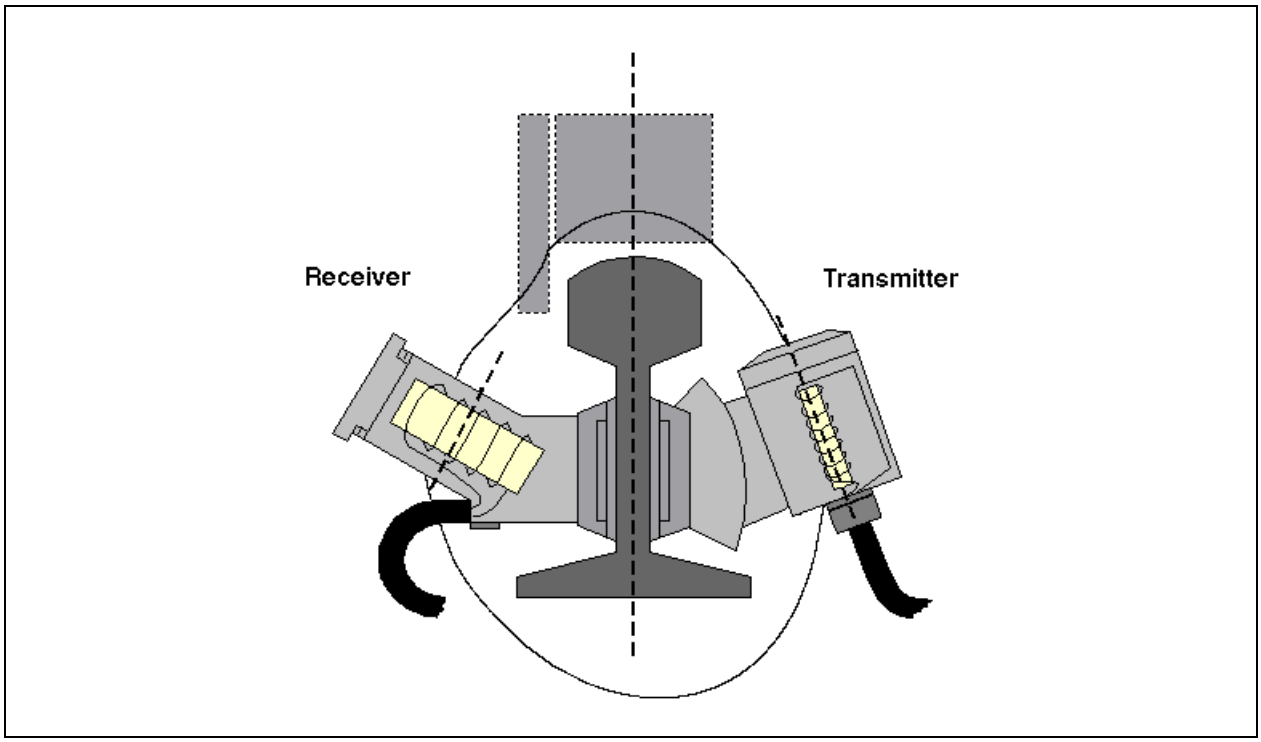


Figure 2.4 Wheel moving away from count head

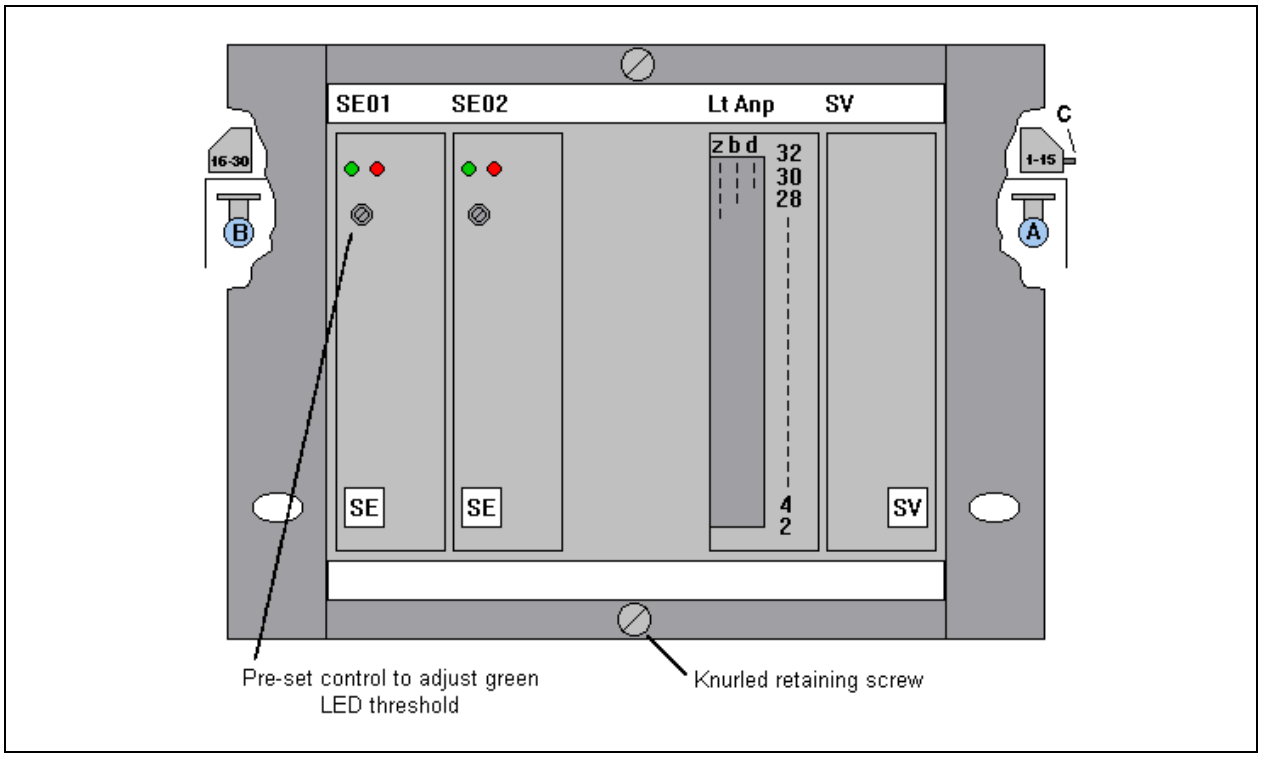
As the wheel moves away from the count head its influence on the magnetic field decreases and the field returns to normal. The voltage in the receiver is now back in phase with the transmitter. The junction box receives the message, 'No wheel over count head'.

THE ELECTRONIC JUNCTION BOX [EAK30]

Layout of the Electronic Junction Box [EAK] equipment

The EAK consists of an enclosed framework which houses four plug-in printed circuit cards and two blocks of screw terminals. The cards and their positions in the rack are labelled.

Figure 2.5 The Electronic Junction Box EAK 30



Key:

A Lightning protection for connection 1 and 2, 3 and 4, 5 and 6.

B Lightning protection for connections 16 and 17, 19 and 20, 21 and 22, 24 and 25.

C 180ohm resistor fitted when this connection is not used. (0.22μF capacitor fitted for the Voice Frequency version,ie.Type D)

1 - 6 Screw terminals for line pair(s) and local power supply (if provided) 7 – 15 Spares

16 - 25 Screw terminals for count head connections 26 – 30 Spares

SE 01 Transmitter/receiver card for SK1

SE02 Transmitter/receiver card for SK2

LtAnp Cable adaptor card

SV Power supply card

gn Green LED

rd Red LED

EAK Card Functions

Power Supply Card [SV]

This card is supplied with a nominal 60 volts d.c. from the Cable Adaptor card and converts this voltage into two stabilised supplies for use by the SE cards. These should be between 22 and 25 volts.

Transmitter/Receiver Cards [SE]

First, note that these two cards are identical, but they generate different frequencies because different connections are used on the edge connectors. Each card receives its own 24 volts supply from the SV card.

SE01 card generates the 30 kHz frequency to drive SK1 Tx. It also detects and rectifies the waveform produced by SK1 Rx. This d.c. waveform (MESSAB 1) is then passed to the Cable Adaptor Card.

SE02 card generates the 29kHz frequency to drive SK2 Tx. It also detects and rectifies the waveform produced by SK2 Rx. This d.c. waveform (MESSAB 2) is then passed to the Cable Adaptor Card.

Each SE card has two LED indications. Their meanings, when lit, are:

GREEN The rectified Rx voltage is positive and at a workable level. This should therefore be lit while no wheel is present. The threshold of this indication is adjustable by means of the small recessed screw (access through the front cover), but it need NOT be set up for the system to work properly. However, see notes on alignment and adjustment of the count heads in Section 4.

With the voice frequency version (ie.type D), the green indication flashes and there is no adjustment.

RED The rectified receiver voltage is zero or negative. In normal operation, this indicates the presence of a wheel. However, with no wheel present, it also illuminates if the connections to the Tx coil or Rx coil are reversed or disconnected.

Cable Adaptor Card [LtAnp]

This card receives a nominal 60 volt d.c. supply from the line pair, or locally, and passes it to the Power Supply Card [SV]. It has two oscillators whose frequencies are 5.06kHz and 4.15 kHz (2.52 and 2.04kHz respectively with voice frequency systems). The d.c. waveforms (MESSABs) produced by the two SE cards are used to switch these oscillators off and on.

There is a third oscillator (not provided in the voice frequency system) which monitors the Green LEDs. It plays no part whatever in the detection or counting of wheels.

A multi-pin socket is mounted on the front of this card to enable the Manufacturer’s test set or a test box (currently made by Electronic Systems Section, Bombardier Transportation, Reading) to be connected. The use of either of these allows voltage measurements, internal to the EAK system, to be taken.

Operation

The EAK receives a nominal 60 volts d.c. (55 - 115V) over the line pair connection from the evaluator or locally (see Power Supplies). This voltage is used to produce two 24 volt d.c. supplies which drive the two SE-cards.

One of these cards produces a.c. at 30 kHz, the other produces a.c. at 29kHz. The 30kHz frequency is applied to the Tx coil of SK1; the 29kHz is applied to the Tx coil of SK2. These frequencies were chosen by the manufacturers so that the count heads could be used on railways where vehicles are fitted with magnetic rail brakes (eg.trams).

The Rx coil of SK1 produces an output at 30 kHz and the receiver coil of SK2 produces an output at 29kHz. These frequencies cannot be sent to the evaluator because it would not be able to utilise them. Two lower frequencies which can be transmitted over longer distances more efficiently have to be sent to the evaluator. These frequencies are 5.06kHz and 4.15kHz (or 2.52kHz and 2.04kHz for voice frequency systems).

These lower frequencies are controlled in the following manner: The two receiver output voltages are phase-sensitive-rectified which means that whilst a receiver voltage is in phase with its transmitter voltage, the result is a positive d.c. voltage referred to as MESSAB, but when the phase of voltage is reversed, due to the presence of a wheel, the result is a negative MESSAB voltage. When the MESSAB voltage associated with SK1 is negative, this cuts off the 5.06kHz frequency transmitted to the evaluator. Likewise, when the MESSAB voltage associated with SK2 is negative, the 4.15kHz frequency is cut off.

This means that the evaluator receives both the 5.06kHz and 4.15kHz frequencies with no wheel present, but loses and regains each one whenever a wheel passes over the detection point. Depending on which frequency is lost first, and whether the associated evaluator input set is configured as a count in or a count out, ‘1’ will either be added to or subtracted from the total count as appropriate.

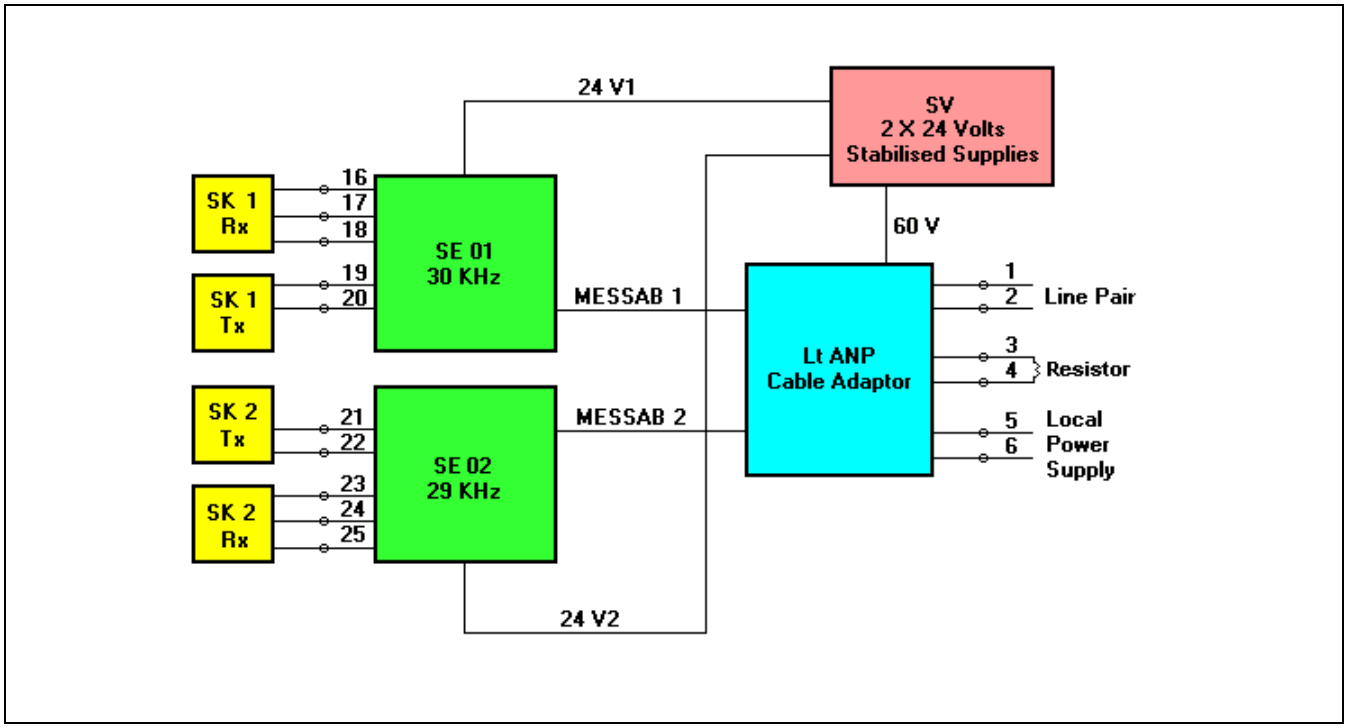


Figure 2.6 EAK 30 block schematic

Terminations in the EAK

1	'a' leg of the line pair from/to the evaluator
2	'b' leg of the line pair from/to the evaluator
3*	'a' leg of line pair from/to a second evaluator
4*	'b' leg of line pair from/to a second evaluator
5	+60 volt d.c. (Used only when the supply is obtained locally.)
6	-60 volt d.c. (Used only when the supply is obtained locally.)
7 to 15	Spare
16	SK1 Rx (black)
17	SK1 Rx (clear)
18	SK1 Rx cable screen (small blue)
19	SK1 Tx (clear)
20	SK1 Tx (black)
21	SK2 Tx (black)
22	SK2 Tx (clear)
23	SK2 Rx cable screen (small blue)
24	SK2 Rx (clear)
25	SK2 Rx (black)
26 to 30	Spare.

* These terminations are used when the detection point applies to two adjacent axle counter sections and the evaluators for the two sections are in different equipment rooms. Otherwise, a 180-ohm resistor is connected across them to act as a dummy load. In voice frequency systems, a 0.22μF capacitor is used instead of the resistor.

Note the colours used on terminations 16 - 25 will vary with the version of the system.

PULSES AND PHASES PART ONE

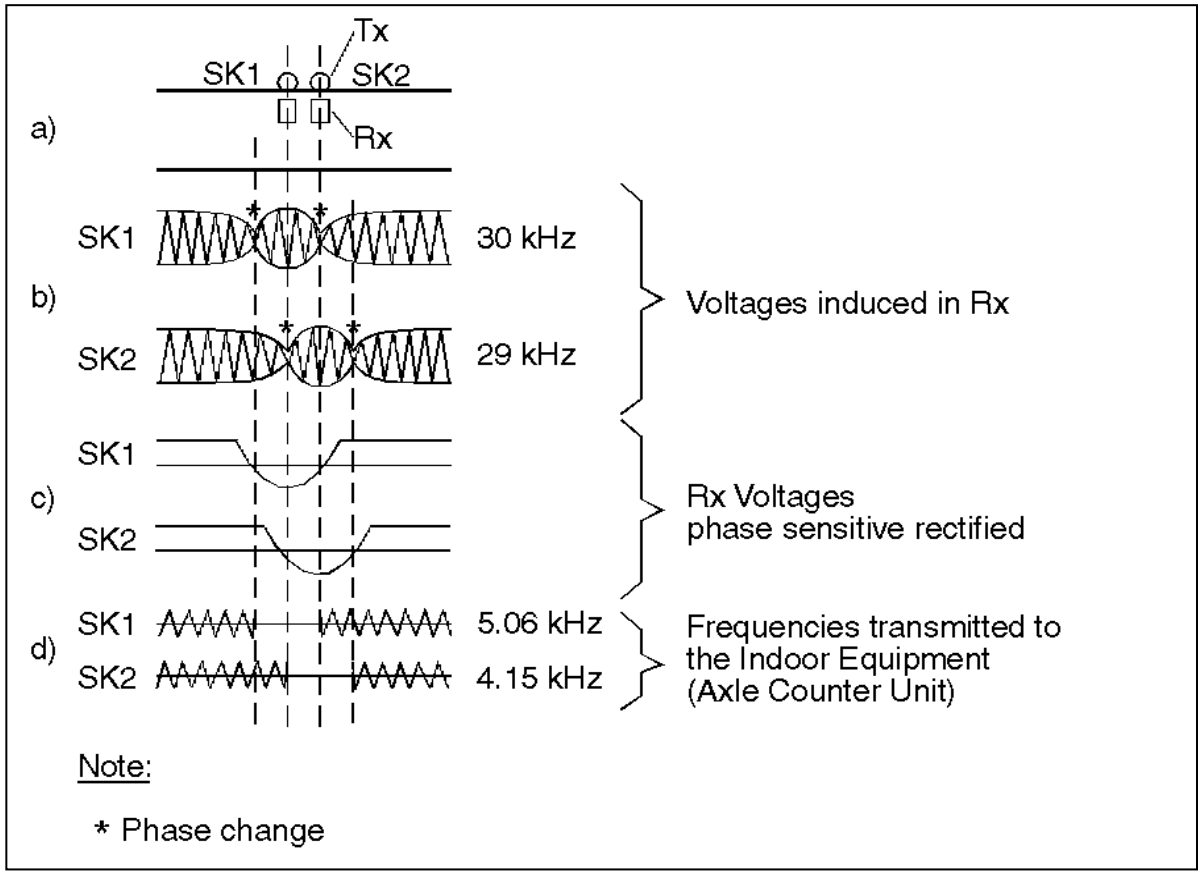


Figure 2.7 Pulses and phases for the outdoor equipment

a) SK1 and SK2 shown on a cess rail, normal direction of travel Left to Right.

b) 30kHz induced in SK1 receiver and 29kHz induced in SK2 receiver. Note how each receiver waveform is influenced by a wheel passing; first a collapse, then reversed phase, then another collapse, and finally back to normal in-phase voltage.

c) The receiver voltages are phase-sensitive rectified. Therefore each d.c. voltage (MESSAB) goes negative when a wheel passes.

d) 5.06kHz frequency cut off to the evaluator, followed by 4.15kHz frequency cut off, then 5.06kHz switched on, followed by 4.15kHz switched on.

Axle Counters FLMF
Section 3
The Indoor Equipment

THE INDOOR EQUIPMENT

This section looks at the structure and operation of the main indoor equipment. It begins with the power supplies, the signalling relays and reset plunger and then moves onto the detail of the evaluator rack and cards.

On completion of this section you will be able to:

1. Describe the structure and function of the evaluator.

2. Describe how the evaluator, relays, reset plunger and power supplies are linked into the system.

POWER SUPPLIES

The axle counter circuits are usually supplied with power from a special set of 60 volt d.c. busbars. These are fed from a 60 volt power supply. This may be provided by a transformer/rectifier and lead acid or nife cells or it may come from a BAUCH or DUVINE power supply unit which contains sealed secondary cells. This arrangement is necessary to prevent loss of power to equipment during generator tests or other interruptions to the main power supply. Should the power to the evaluator ever be lost and then reinstated, the equipment will remain in the fault condition, which is “track occupied” as far as the rest of the signalling equipment and the signalman are concerned. This fault condition is retained until the Reset Procedure is carried out. This is necessary because axles may have entered or left the axle counter section during the power supply loss.

SIGNALLING RELAYS AND THE RESET PLUNGER

Near to the evaluator rack is a twin 50 volt relay, the TZR/TSR. Also located nearby is the TPR, the contacts of which are used conventionally. The other item in close proximity is the technician’s reset plunger and green lamp which lights when the TZR is down and the TSR is up, i.e. the condition which allows the TPR to energise.

THE EVALUATOR

Layout

The evaluator rack is a framework able to hold up to a total of 24 plug-in cards. Access to these is from the front. Each card is mounted vertically and connected to the rest of the system through an edge connector at the rear. A plastic clip prevents the card from sliding forward. Each card is identified by a series of letters (e.g. [WDH], [GRDFR], etc.). which also appear on the rack frame immediately above the card slot.

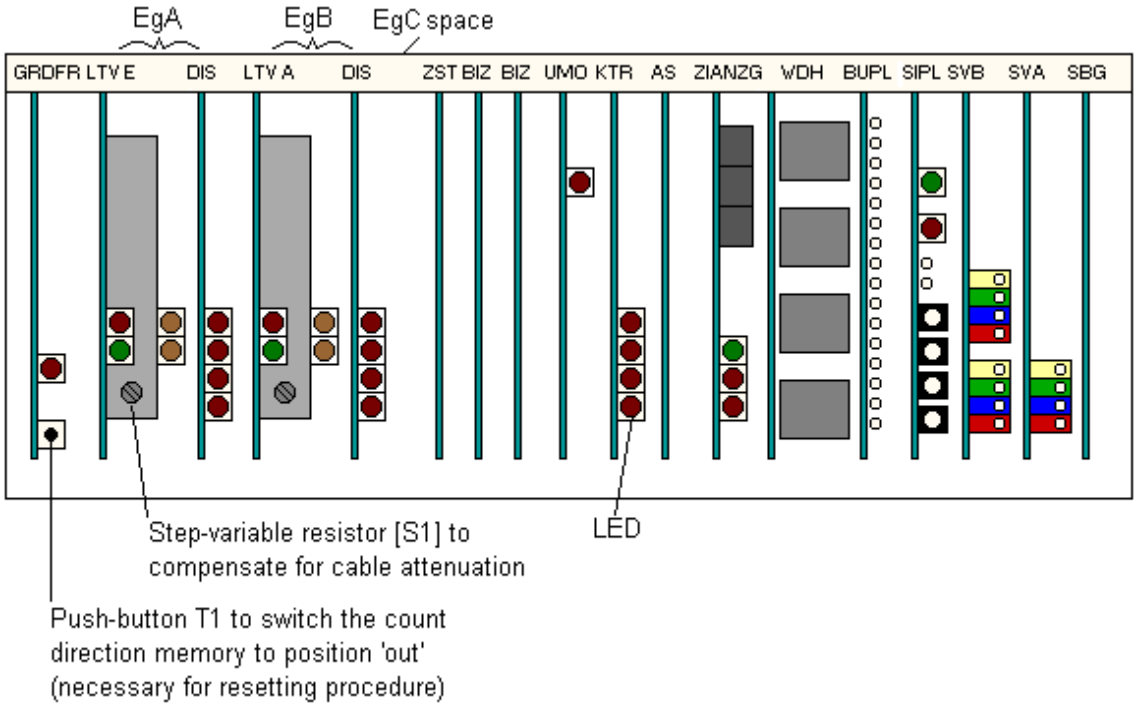


Figure 3.1 The evaluator rack and cards

Connections are made to the evaluator rack via a pair of 20 x 7 solder-lug tag-blocks on the rear of the subrack. These are covered when the system is in use. The tags have a prefix referring to the axle counter section I.D. e.g. AB -/--

NOTE: The left hand number refers to the tag block and the right-hand number to the tag.

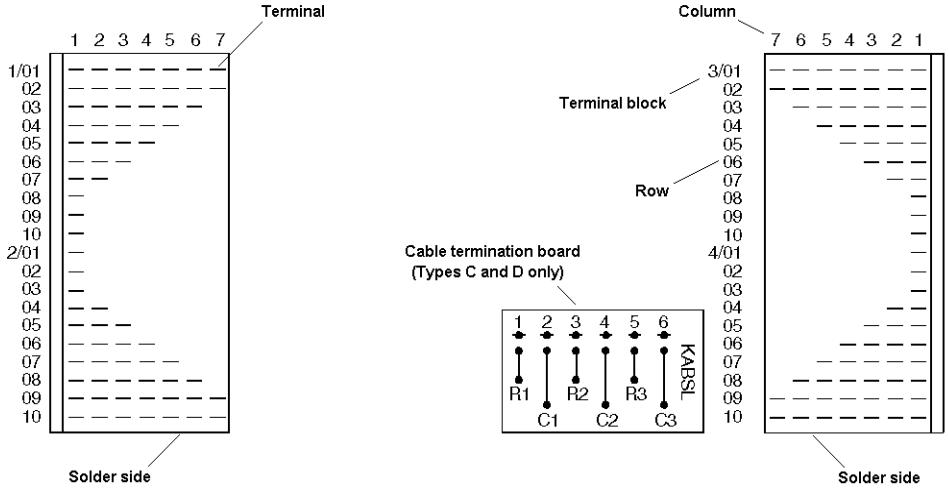


Figure 3.2 Rear of evaluator subrack terminal blocks

Inputs

Power for each evaluator rack is usually supplied by a special pair of 60 volt d.c. busbars, via solder lug connections. The line pair from each detection point is connected to solder lug connections on the evaluator rack via a lightning protection unit (LPU) on the cable rack.

When the technician's reset plunger is pressed, a feed from the 60 volt power supply is applied to the appropriate part of the evaluator via a solder lug connection (see figure 3.2.).

Outputs

The evaluator drives the TZR/TSR relay via its own relay interface card [WDH] and the solder lug connections at the rear of the evaluator (figure 3.2). The TPR and green lamp operate when the TZR is Down and the TSR is Up.

OPERATION OF THE EVALUATOR - CARD FUNCTIONS

The Reset Inhibit Card [GRDFR]

The GRDFR card prevents the technician's reset plunger being effective if the counter last processed a *count in*. On the card there is a relay, K1. The Reset Relay, AzT, on the WDH card, requires K1 Up to energise. K1 is energised either by a **count out** or, **during fault-finding ONLY**, by pressing button T1 on the GRDFR card. This button converts an existing *count in* to a *count out*. (Note that GRDFR is a simplified version of the FRMKTR card used in Type A and B evaluators.)

The Input Set [Eg] ([LTV-E], [LTV-A] and [DIS])

The incoming signals on each detection point line pair are first processed in an input set of cards. Depending on whether the detection point is a **count in** or a **count out**, its [LTV] card will be [LTV-E] or [LTV-A] . The evaluator can hold up to three sets of input cards, known as [EgA], [EgB] and [EgC].. Therefore one evaluator can accommodate up to three detection points, i.e. for an axle counter section through a single turnout. Note that the LTV cards are ***not*** identical, so their type numbers must be checked when replacing them.

In the LTV card, the two incoming wheel detection frequencies, 5.06kHz and 4.15kHz, are first separated by filter circuits and then amplified. The rotary switch, S1, adjusts the gain to overcome line loss. The signals are then rectified to produce a 5 volt d.c. logic voltage. Loss of this voltage is called a wheel detection pulse (see figures 3.3 and 3.10). The [DIS] card recognises the order in which the wheel detection pulses are formed, so it can **DIS**criminate between a **count in** and a **count out**.

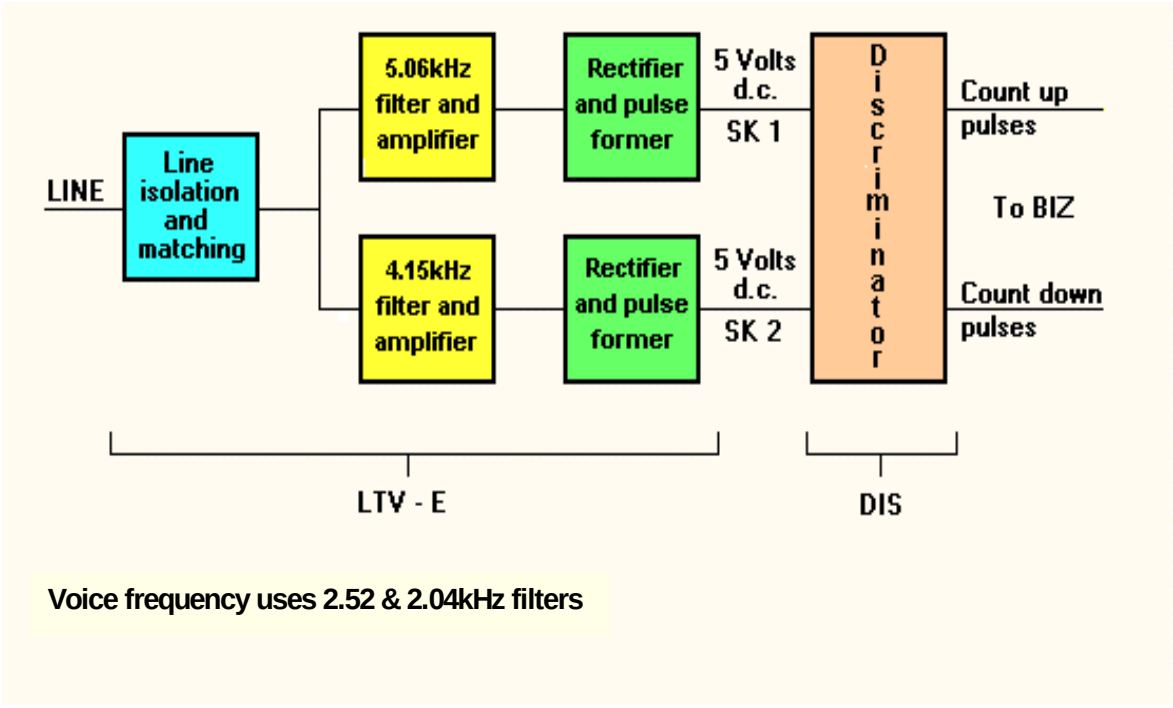


Figure 3.3 “Count in” input set block schematic

Each Input Set comprises a pair of cards connected together (LTV) plus a DIS-card. The LTV carries out line matching, filtering, amplifying, rectifying and pulse-forming. Note that LTVs are different for the count-in and count-out ends, but their “keyway” configurations are the same. For the Type C system the LTV-E (count-in) is part No. 00090 and the LTV-A (count-out) is part No. 000100. each LTV card sends two 5v D.C. “Wheel pulses” to the DIS-card. Also, each LTV-card contains a means of adjusting the Gains on the two amplifiers.

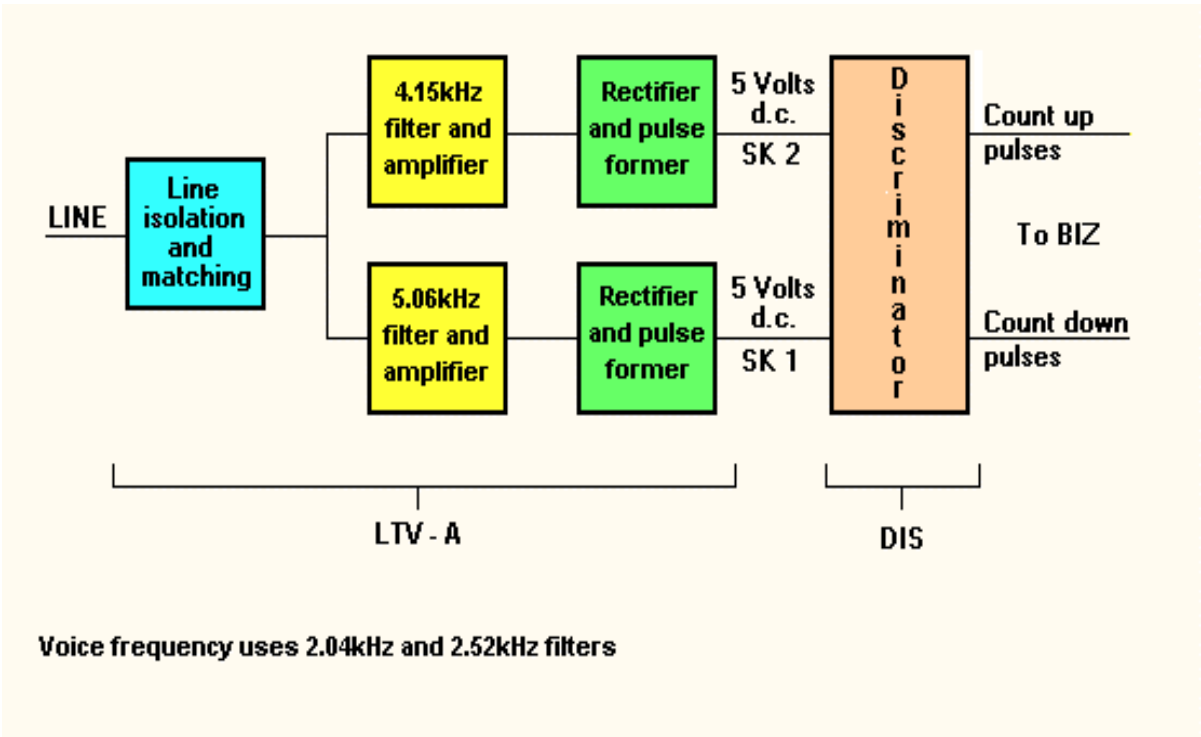


Figure 3.4 “Count out” input set block schematic

The Count Cards 2 x [BIZ], [KTR] and [ZST]

The following cards form the counter and its associated support functions: 2 x BIZ, KTR and ZST. The binary counter itself is located on the BIZ cards. The counter is directly influenced by the actions of the input sets, being stepped ***up*** or ***down*** as necessary. The counter ultimately drives the GB relay on the WDH card. The GB relay is normally up when the counter is ***NOT*** at zero, i.e. Track Occupied.

Closed Loop Supervision	
Counting Supervision	Dynamically checks the operation of both the counter and the input sets. See LEDs G24 and G26 on the KTR card and G2 on the ZIANZG card for status.
Track Clear Supervision	Utilises information from both the counter *and* the inputs sets to determine whether the section is clear or not. See LEDs G25 and G27 on the KTR card and G3 on the ZIANZG card for status.
Power Supply Supervision	Monitors the system power supply levels.
Input Set Monitoring	Checks the input set threshold (5V) levels.

Various other elements are also associated with the counting process and together form the 'Closed Loop Supervision' function.

Together, the elements of the Closed Loop Supervision function ultimately control the GF relay on the WDH card. *Note* the GF relay will be UP with the Section Clear and No System Faults.

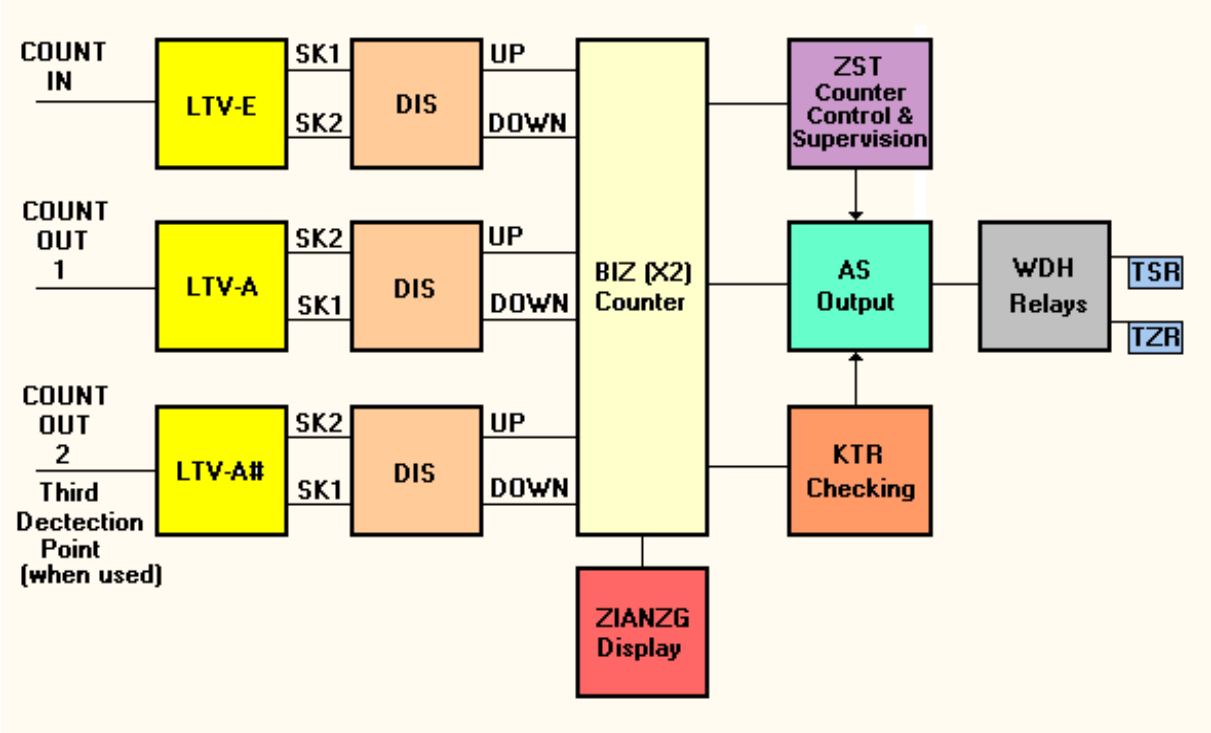
The Count Indicator [ZIANZG]

The [ZIANZG] card contains a three-digit seven-segment counter display, plus three LED indications (see figure 3.1). The card need not be plugged in for the system to operate. It has two solder lugs near the front for testing the counter display. When these are shorted together, all segments should light up displaying 888.

If the counter receives a *count out* without a previous *count in*, perhaps as a result of a fault or during fault-finding, a count down from 512 will occur. This is because a nine-stage binary counter is used in the standard system and there is no such thing as a negative count. (Note that 512 is never displayed because this is equivalent to Zero. The first axle out, <u>without</u> a count-in, will cause 511 to be displayed.)

Sometimes, only one ZIANZG card is provided at each site where evaluators are present. It should be left in the system where last used, not stored unpowered.

Figure 3.5 Evaluator block schematic



The Interface Relays [WDH]

The [WDH] card contains four open-style relays:

1. [GF] relay, energised when the electronic circuits detect Section Clear, assuming there is no fault with the electronics.

2. [GB] relay, energised when the electronic circuits detect Section Occupied OR during certain electronic fault conditions.

3. [GBP] relay, energised when the true Track Occupied condition exists, i.e. [GF] Down and [GB] up, or when Track Occupied is simulated by Reset [AZT].

4. [AzT] relay, energised when the technician's reset plunger is pressed, except when a reset is unnecessary, i.e. GF Up and GB Down.

The table overleaf shows the consequences of the conditions of the WDH card relays on the status of the axle counter.

Indication	GF	GB	GBP
Track Clear	↑	↓	↓
Track Occupied	↓	↑	↑
Fault	↑	↑	↓
Fault	↓	↓	↓
Fault	↓	↓	↑

↓ = Relay dropped away ↑ = Relay picked up

The indication “Track Occupied” is supplied to the signalling system via the signalling relays TSR, TZR and TPR. The WDH card also contains the relay AzT which is required for resetting the axle counter to the normal condition.

In normal operation, the GB relay will control the TZR (Track Occupied Relay) and the GF will control the TSR (Track Clear Relay). The WDH card has a fuse. If this fuse is blown, the GF and the GB relays cannot operate. It does not affect GBP and AzT relays.

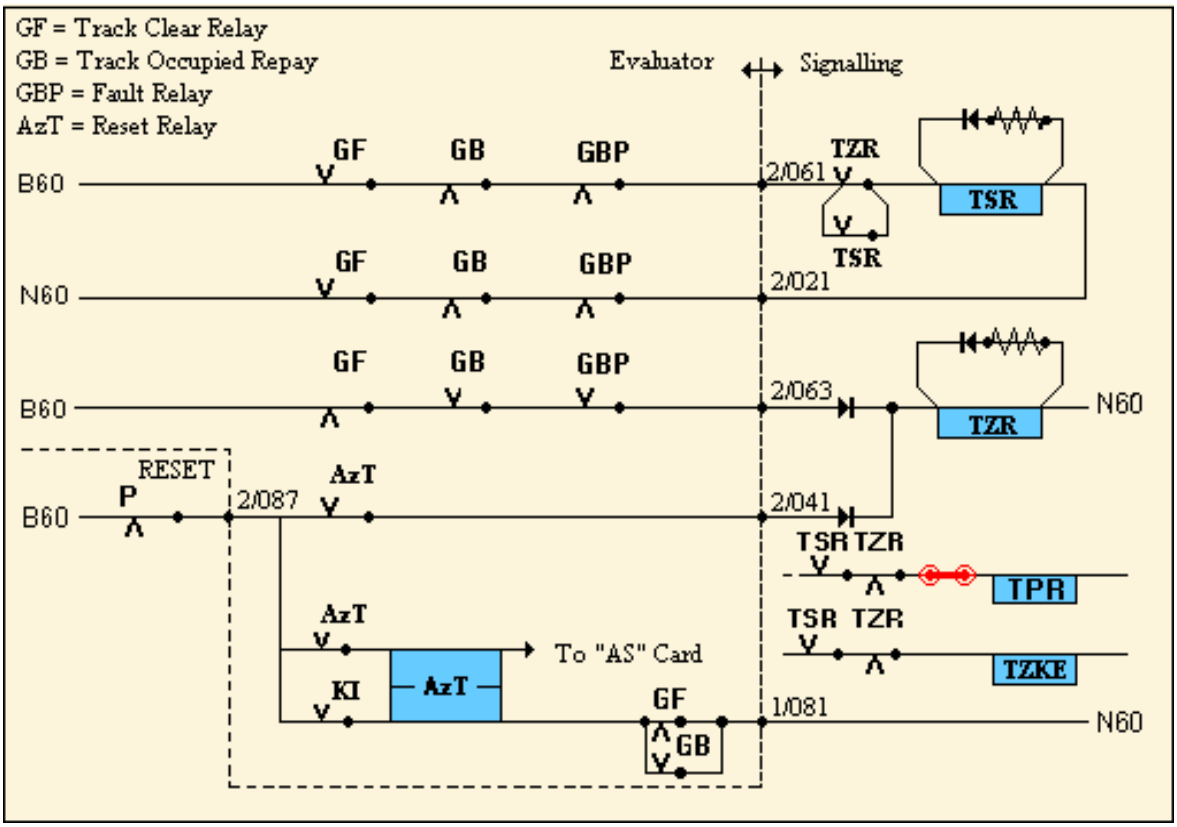


Figure 3.6 Interface and signalling relays

Note: KI is a relay on the GRDFR card. This is energised either by a train counting out or by the Technician pressing the button the GRDFR card.

The Test Card [BUPL]

The BUPL card has a series of Test Jacks along its front edge. The 4mm sized jacks are the most important for first-line fault-finding.

The Fuse Card [SIPL]

The SIPL card houses four fuses:

Si 1, Si 2 and Si 3 feed each of three detection points via their line pairs. Si 4 feeds the power cards SVA and SVB with 60v d.c.

The SIPL card also has two indications and two test jacks for voltage measurements.

The Power Supply Cards [SVA], [SVB], [SBG]

SVA card and SVB card produce 60, 70, 80 or 90V, selected by a 2mm plug for each detection point supply. However, the *single rail* axle counter system is usually set up to supply 90V to all its detection point line pairs. Therefore a plug should ONLY be in the RED socket for each of the three sets of voltage selection sockets, one in SVA for detection point A, one in the top set of SVB for detection point B and one in the bottom set of SVB for detection point C. All three should be fitted even if not used. NEVER INSERT TWO PLUGS IN ANY ONE SET OF FOUR SOCKETS. THIS WILL SHORT THE TRANSFORMER – EXPENSIVE!

Strap connections on the evaluator rack rear tag-blocks determine whether or not a detection point is supplied from the evaluator.

The SVB card also has a fuse which supplies 20V d.c. to the evaluator rack electronics. As this supply comes via the SVA card, it relies on both the SVA and SVB being operational.

The SBG card contains filters associated with the incoming 60V d.c. supply and the 20V d.c. supply for the evaluator rack.

FUSES

All fuses in the system are indicating type and must be placed in their fuseholders with the coloured “pip” facing forwards.

In the evaluator there are the following fuses:

1. SIPL Card

Si 1	125mA, 90V d.c. supply (selected) to Detection Point A [ZP 1].	(Purple)
Si 2	125mA, 90V d.c. supply (selected) to Detection Point B [ZP 2].	
Si 3	125mA, 90V d.c. supply (selected) to Detection Point C [ZP 3].	
Si 4	1 amp, 60V d.c. input from busbar to all equipment. (Red)	

2. SVB Card

Si 2 630mA (*at **rear** of card*), 20V d.c. supply to all electronics. (Green)

3. WDH Card

Si 1 1 amp, 60V d.c. supply to GF and GB relays. (Red)

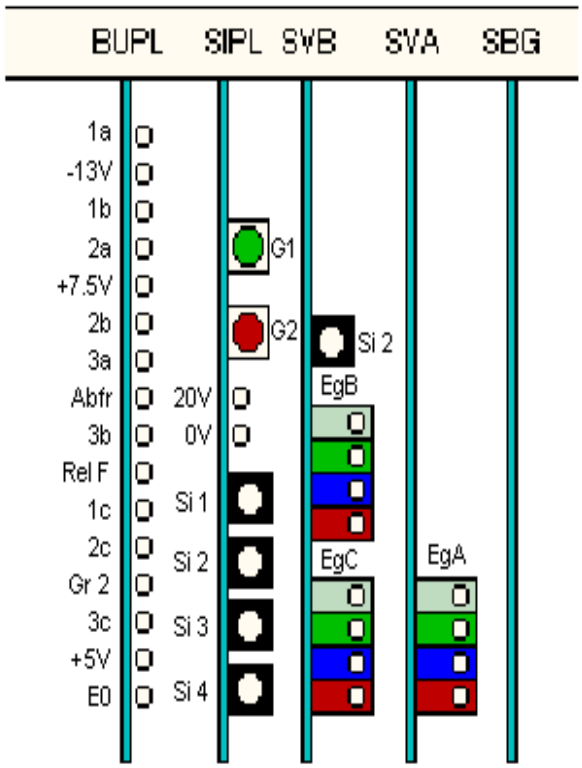


Figure 3.7 The fuses

EVALUATOR INDICATIONS

Several of the cards are equipped with LED indications. A brief explanation of the meaning of each one is shown below.

Card	LED	Indication when lit
GRDFR	+ G10 (Red)	Last counter action was Count Up, Reset NOT possible. (Note 5 second delay for LED to light.)
LTV-E	H1 (Red) + *H2 (Grn)	A green LED on SE-card at Detection Point is out. LED will not light if a wheel is present, ie. a Yellow is out. (Note 15 second delay for LED to light.) Green LEDs on SE-cards are lit. (Not necessary for system to work, but desirable for consistency.)
LTV-E	*H1 (Yell) *H2 (Yell)	a.c. from line pair for SK1 (5.06kHz) is present. a.c. from line pair for SK2 (4.15kHz) is present.
LTV-A	H1 (Red) + *H2 (Grn)	See LTV-E
LTV-A	*H1 (Yell) *H2 (Yell)	a.c. from line pair for SK**2** (4.15kHz) is present. a.c. from line pair for SK**1** (5.06kHz) is present.
DIS	G1 (Red) G2 (Red) G3 (Red) G4 (Red)	a.c. from line pair for SK1 (LTV-E) or SK**2** (LTV-A) is low. a.c. from line pair for SK2 (LTV-E) or SK**1** (LTV-A) is low. Wheel presence has been registered in store 1. Wheel presence has been registered in store 2.
UMO	G12 (Red)	Power for the evaluator has been lost and re-instated. This indication 'sticks' in the fault state, so the system has to be Reset in accordance with the Reset Instructions.
KTR	G24 (red) G26 (Red) + G25 (Red) + G27 (Red)	Counting fault in counting check store 1. Counting fault in counting check store 2. Store 1 is set, which means the track is NOT clear. Store 2 is set, which means the track is NOT clear.
ZIANZG	*G1 (Red) + *G2 (Red) *G3 (Grn)	Last Counter action was Count Down. Count supervision successful. Counter at zero and system functioning correctly. **Note:** This card does not need to be plugged in for the system to operate.
SIPL	*G1 (Grn) + G2 (Red)	Track Clear. Track Occupied or Fault. **Note:** these LEDs monitor ONLY the electronics, NOT the WDH relays.

* These LEDs should be lit if the Section is Clear and the system is functioning normally.

\+ These LEDs should be lit if the Section is Occupied, no wheel is passing over a Detection Point and the system is functioning normally.

The green lamp near the Reset Plunger monitors the TSR and TZR relays and is lit when the Section Clear condition exists (i.e. TSR UP and TZR DOWN). It is fed over different contacts from those which feed the TPR Relay.

The following diagrams show the indications for the AZL 70/30 high frequency evaluator N.B. cards without LEDs are not shown.

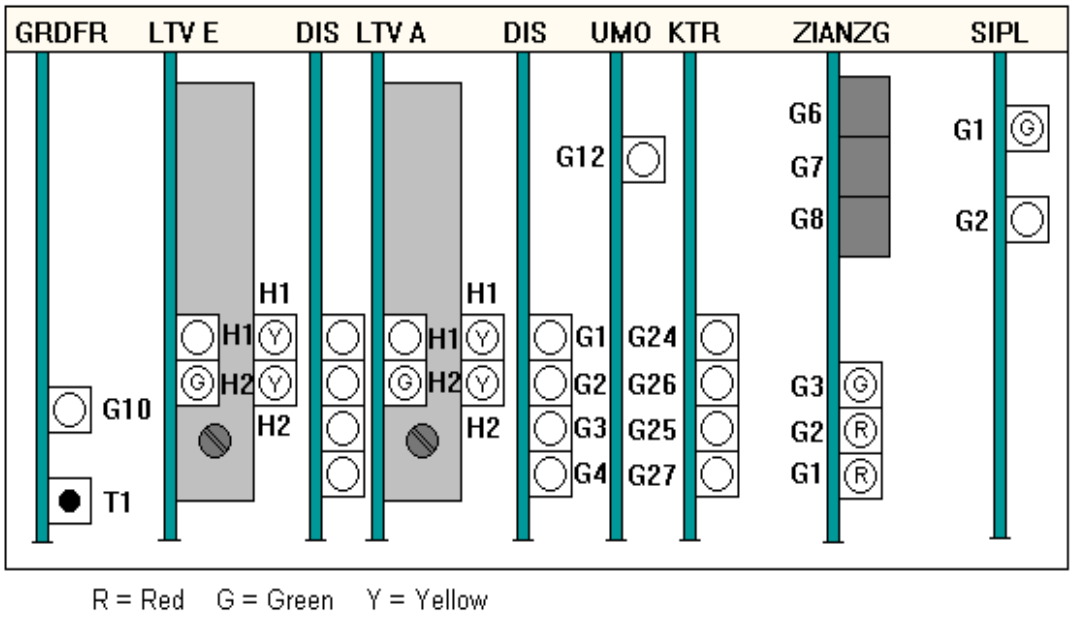


Figure 3.8 System normal - no train in section

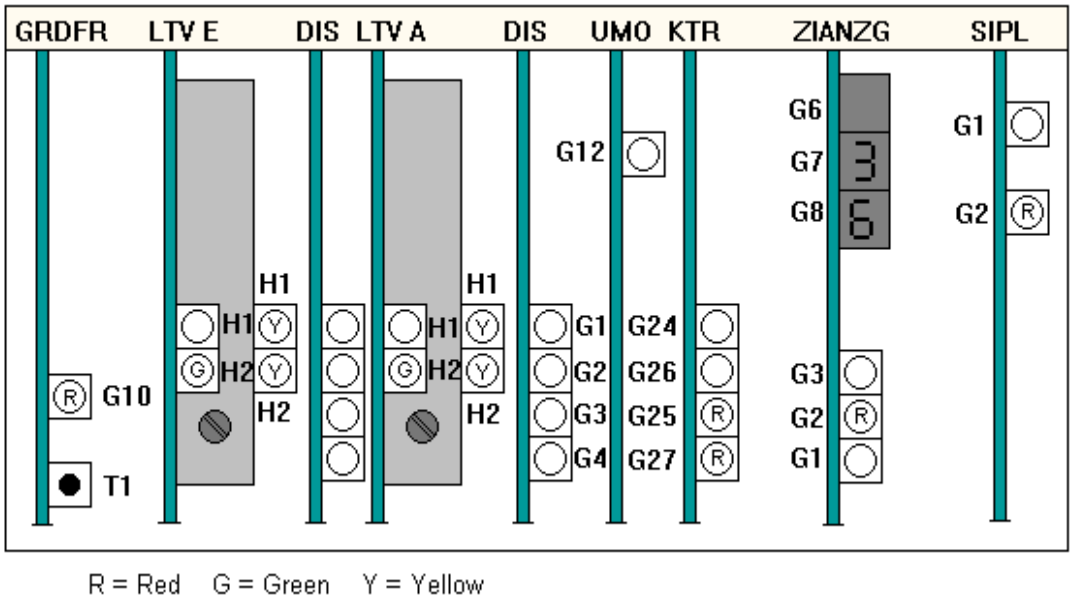


Figure 3.9 System normal - train in section, 36 axles counted in; none out

PULSES AND PHASES PART TWO

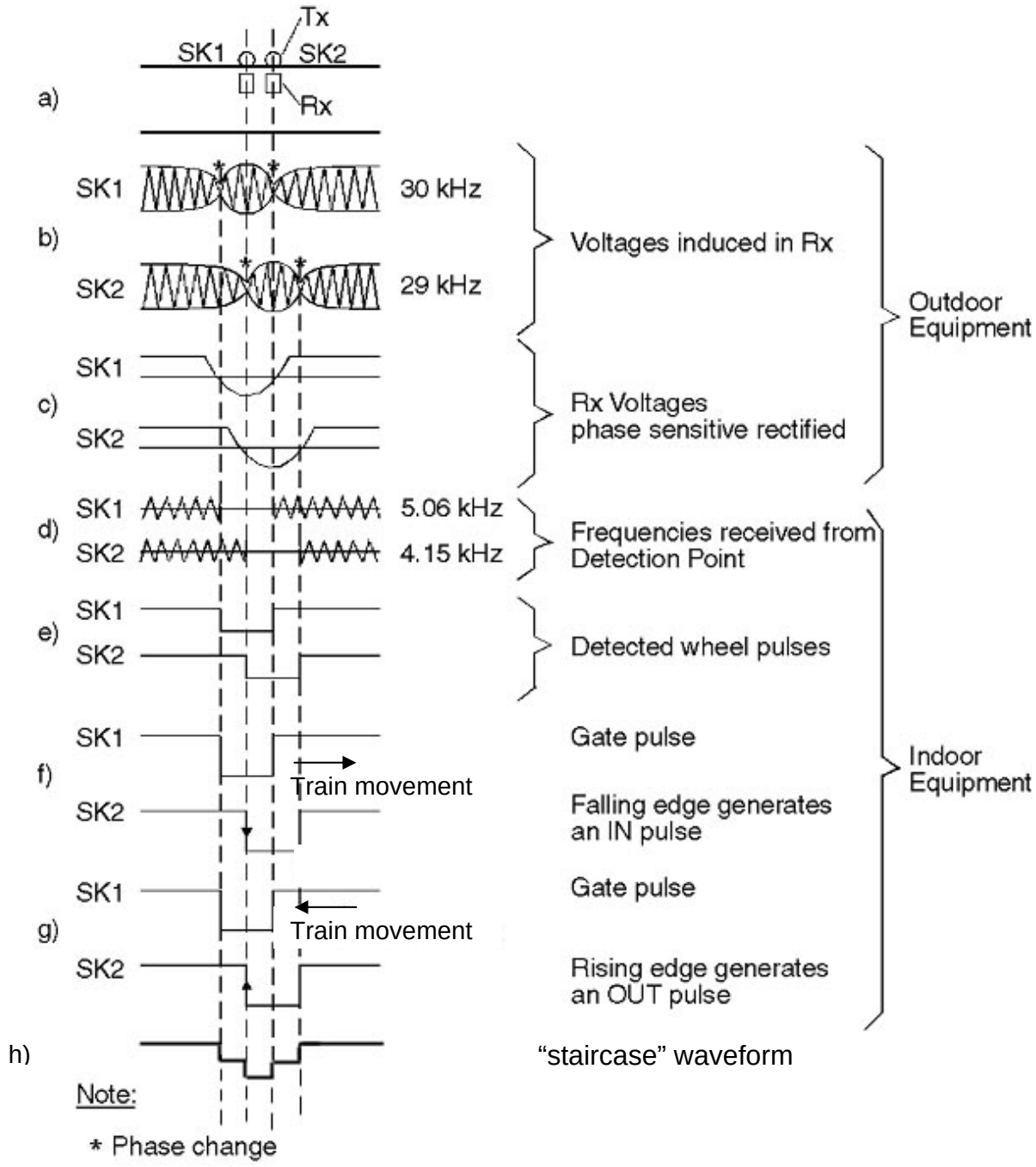


Figure 3.10 Waveforms for Count-IN end & Evaluator

Axle Counters FLMF
Section 4
Maintenance

MAINTENANCE

This section starts by introducing the test equipment used in maintenance and fault-finding. It then looks at maintenance routines and the adjustment of the various components.

On completion of this section you will be able to:

1. Select and use the test equipment correctly.
2. Perform maintenance on an AzL 70/30 system in accordance with the relevant maintenance procedures.
3. Compare the results with previous results and acceptable values.
4. Take any necessary corrective action.
5. Record the values.
6. Report accurately to the supervisor items requiring attention which are outside the technician's scope.

THE TEST EQUIPMENT

The tools required for maintenance are:

Small and large screwdrivers
Small adjustable spanner, 4-6" handle
Socket set (13mm for count head adjustment)
Digital multimeter (Fluke 87)
Lineside test-switch box with **short** twisted leads
SEL dummy wheel
Record sheets.

Useful but not essential:

Counting indicator board (ZIANZG)
Counting test unit
Digital storage oscilloscope
Short test strap 4mm plug - 4 mm plug.

Voltage Readings

Voltage measurements may be made during routine maintenance and are *ESSENTIAL* during fault diagnosis**.** The voltage readings are taken with a Fluke 87 or equivalent.

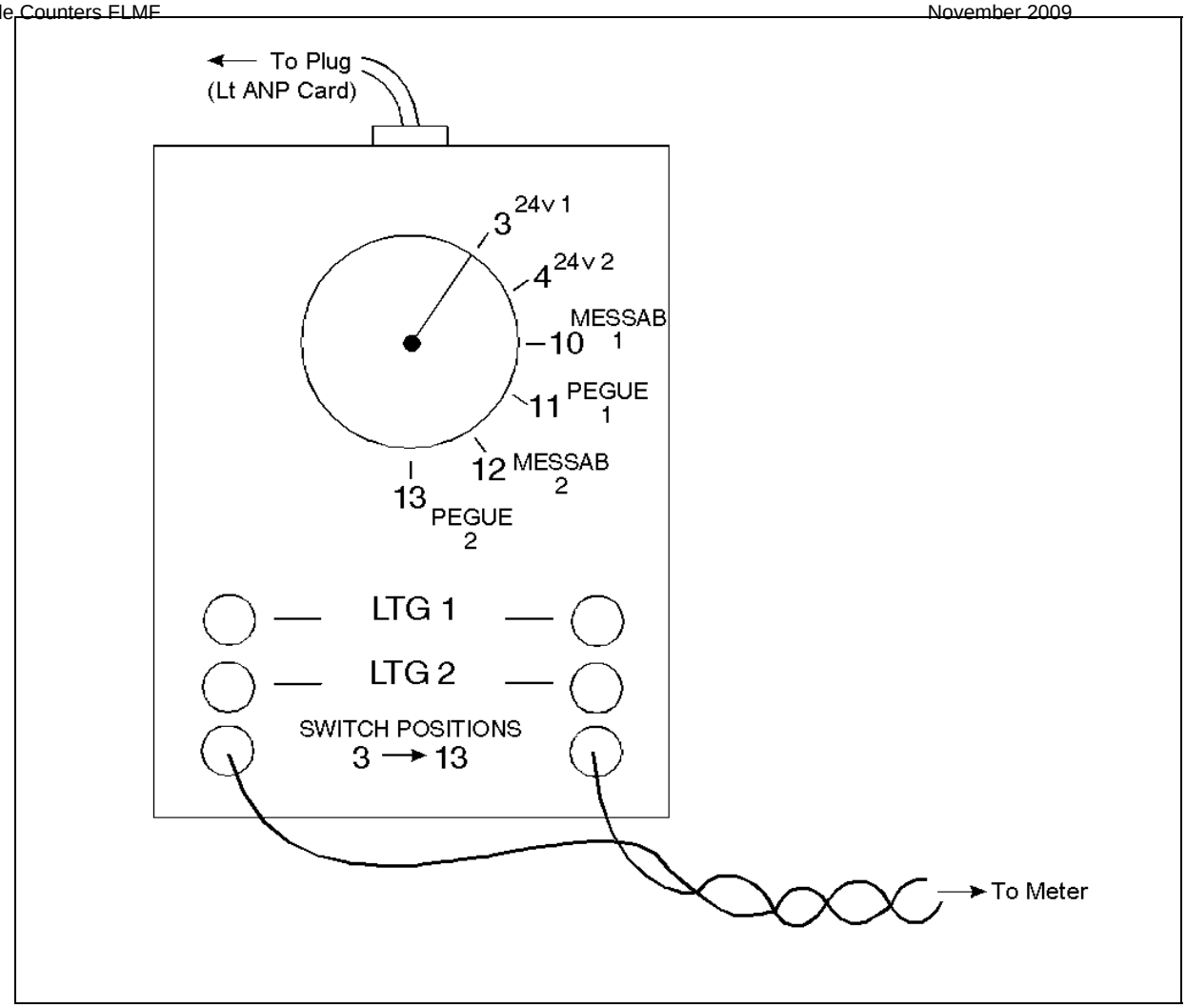


Figure 4.1 The Test Box

The Test Box

For the purpose of testing the detection point equipment, the manufacturers provide a Test Set. Because this is quite large and has many more settings than are required for first-line maintenance and faulting, some teams use the smaller Test Box devised and built by the Electronic Systems Section at Reading in B.R. days. This can be plugged into the LtAnp card socket instead of the Test Set. A meter, Fluke 87 or equivalent, is plugged into the Test Box. For ease of cross-referencing with the Manufacturer's Manual, the Manufacturer's Test Set selector switch numbers are referred to on the Test Box, even though only a few are used, i.e. 3, 4, 10, 11, 12, 13. The others (1, 2, 5 to 9 and 14 to 24) are not included.

Dummy Wheel

To simulate a wheel on a single rail axle counter, a metal plate with three plastic brackets is used. For AzL 70/30 systems, the bottom edge of the plate must be adjusted to 40mm below the rail top and the plate must be perpendicular to the rail. This corresponds to the influence of a 13 inch diameter wheel, the smallest likely to require detecting.

Two brackets are swung out 90 degrees to the plate and rest on the rail top. The third bracket is flat and parallel to the steel plate. It holds the steel plate in the correct position by resting on either one of the receiver coils to influence only one count head, or across both receiver coils to influence both count heads simultaneously. If it is required to simulate several wheels passing over the count

heads, loosen the knurled nut and raise the flat bracket so that it is clear of the receiver coils while the device is passed over the heads.

MAINTENANCE PROCEDURES

Maintenance procedures are contained within the Signalling Maintenance Specification (SMS) AX12**.**

An example of maintenance procedures can be found in the Appendix 2. The procedures are divided into two: Maintenance Visit and Extended Maintenance.

Overleaf is some information which you will need to use when carrying out maintenance tests**.** Further values are contained in the example Maintenance Procedures.

Test points ('Jacks') for voltage measurements

1. *The stabilised 20 Volt d.c. supply which powers the evaluator circuits can be monitored across the Test Points on the SIPL card labelled '20V' and '0V'. This voltage should be between 19 and 21 Volts. '0V' Test Points are on the BUPL and SIPL cards.*

2. *BUPL card Jack Points and their uses are as follows:*

JACK	***INPUT***	***COUNT HEAD***	***ADJUSTMENT***
1a	*SK**2***	*Count in, Input Set EgA*	*LTV-E card, Adj. S1*
1b	*SK**1***	*Count in, Input set EgA*	*LTV-E card, Adj. S1*
2a	*SK1*	*Count out, Input Set EgB*	*LTV-A card, Adj. S1*
2b	*SK2*	*Count out, Input set EgB*	*LTV-A card, Adj. S1*
3a	*SK1*	**Count out, Input Set EgC*	*LTV-A card, Adj. S1*
3b	*SK2*	**Count out, Input set EgC*	*LTV-A card, Adj. S1*

** Where provided.*

*Beware of the infuriating "cross" at 1a and 1b, being SK**2** and SK**1** respectively!*

Test Points 1c, 2c and 3c are for checking the staircase waveform of each input set (see Manufacturer's Manual).

3. a) *A nominal 5 Volts d.c. should be measured between each of the six BUPL card jack points and '0V', e.g. meter positive on BUPL 1a and meter -ve on '0V' will show the condition of SK2 count head at the* count in *detection point. These voltages should be between 4.7 and 7.0 Volts.*

   b) *If a count head is not functioning correctly, or if a wheel is over the count head at the time of measurement, approximately 1 Volt d.c. will be seen at the appropriate jack.*

   IMPORTANT: If any BUPL card voltage is below 4.7V with no wheel present, check FIRST the appropriate detection point equipment and line pair. DO NOT adjust the gain on the relevant LTV card without good reason.

4. *The a.c. voltage coming in, and the d.c. going out (where appropriate) on the line pair, can be measured conveniently at the lightning protection unit on the cable rack. (The solder lugs on the back of the evaluator are not readily accessible). These readings can be checked against the commissioning results to see if there is any reason to suspect a line fault. However, the a.c. measurement is only indicative because it depends on whether one, two or three frequencies are present on the line pair from the relevant junction box.*

Line level test

1. If the meter is connected to LTG-1 on the test box, it will read the voltage(s) on the line pair connected to terminals 1 and 2. LTG-2 sockets are connected to terminals 3 and 4 so they are used only when the detection point applies to two adjacent axle counter sections with the evaluators in different equipment rooms.
2. There will be several voltages on the line pair:

a) 60V d.c. (nominal value) from the evaluator. If the supply to the detection point is locally obtained (terminals 5 and 6 are used) d.c. volts will still be present because of the way terminals 1 - 6 are linked internally.

b) Three a.c. voltages of frequencies 5.06kHz (SK1), 4.15kHz (SK2) and 2.53kHz (monitor).

3. If it is necessary to test the level of the 5.06kHz voltage, place the dummy wheel over count head 2 so as to cut off 4.15kHz. Similarly, place the dummy wheel over count head 1 to cut off 5.06kHz, leaving 4.15kHz on the line pair.
4. Electrical noise on the line pair can affect the operation of the axle counter system. The noise level must not exceed 100mV a.c. To test for this, place the dummy wheel centrally over both count heads. This will cut off all frequencies going to the line pair. The red LEDs must be lit on the SE cards. Any a.c. noise will be shown on the meter, set to a.c. Volts, connected to LTG-1 sockets on the test box (or LTG-2 if appropriate).

COMPARE THE RESULTS

As with the maintenance of all equipment, you need to check that all the readings which you obtain are acceptable**.** The values need to be checked against the acceptable value ranges**.** You should also compare the values with those obtained on previous maintenance visits to ensure that there is no deterioration**.** If there is a significant change in a value(s), although it is still within acceptable limits, you should inform your supervisor.

If you obtain values outside the acceptable limits, or you have noted a problem from your visual checks, you need to take corrective action**.** If this action is outside your responsibilities you will need to sign the section out of use, inform your supervisor and complete any necessary paperwork.

TAKING CORRECTIVE ACTION

You will need to take action to correct potential failures where the action is within your scope of work. Some examples are:- a problem identified visually with a cable or count head guard, a transmitter misaligned so not sending the correct signals, a card malfunction or a power supply deterioration.

Aligning a Transmitter Head

To ensure effective wheel detection, the count heads at a detection point must be able to detect accurately the presence of a wheel passing over them. With no wheel present, the rectified receiver voltages [MESSABs] are positive, between 55mV and 1000mV. A wheel directly over a count head causes its MESSAB voltage to go negative. For efficient control of the evaluator, it is desirable to arrange for the value of these voltages to be the same. For this reason, the transmitter on each count head can be adjusted up and down on its mounting. Note that the receiver is fixed.

The ideal arrangement is to fix the transmitter head so that MESSAB POSITIVE = MESSAB NEGATIVE. In practice, this is rarely achieved (it largely depends on how accurately the rails have been drilled) but, if the two values are within 25mV of each other, the system is satisfactory.

The suggested procedure for aligning a transmitter head

1. Obtain possession of the axle counter section(s) which you are about to affect. Use the specially provided DISCONNECTION LINK to disconnect the section TPR.
2. Remove the cover on the lineside junction box by removing the four fixing screws carefully and lifting the lid.
3. Plug the test box into the socket on the LtAnp card in the EAK, then switch on the meter, select Volts d.c. and plug the meter into the test box using the red and black sockets marked Switch Positions.
4. Select position 3 and record the reading (22 - 25V). **DO NOT PROCEED FURTHER IF THIS SUPPLY VOLTAGE IS WRONG.** Instead, investigate the cause. (A suggested procedure is: check the incoming supply; substitute the EAK cards - a fault on any one of them may cause this symptom.)
5. Select position 10 and record the reading**.** This is positive MESSAB 1 (+55 to +1000mV).
6. Place the dummy wheel (set at a depth of 40mm) on the rail over count head 1 and record the reading**.** This is negative MESSAB 1 (-55mV to -1000mV). The dummy wheel should be positioned at 90° to the sleeper. Depending on the track cant, this may not necessarily be vertical.
7. Now repeat for other count head using Switch Positions 4 and 12.

   a) If the value of these two voltages (steps 5 and 6) are within 25mV of each other, the Tx coil is in good alignment and the system can be left to work in this condition, but see 9 below.

   b) However, if these MESSAB voltages differ by more than 25mV, slacken the self-locking nuts holding the transmitter against the serrated mount and move the transmitter DOWN one notch to INCREASE the MESSAB +ve voltage and move it UP to DECREASE the MESSAB +ve voltage. Hold the transmitter head very firmly against the serrated mounting block while carrying out steps 5 to 7 again.

8. Ensure that the last measurements taken are with the transmitter nuts fully tight. **IT IS IMPORTANT** to use only the nuts supplied for the job. If lost or worn, replace with the correct spare; they are slightly oval (pinched across two flats) which gives the self locking effect.

   CAUTION:** **if a transmitter drops down to its lowest adjustment, there is a possibility that the count head will not detect small-diameter wheels. Therefore, if you have any doubt about the security of the transmitter nuts &/or bolts, renew accordingly.

9. Once satisfactory MESSAB levels have been obtained, you can check the PEGUE levels. Select switch position 11 (13 for count head 2) and observe the reading. This should be the same as the MESSAB +ve voltage.

Test points 11 and 13 are pre-set reference voltages [PEGUE] which give the facility for monitoring any significant drift in the MESSAB +ve voltages which may have occurred between visits. The green LED on an SE card will go out if the MESSAB +ve voltage is different from the PEGUE voltage by about 20%. The PEGUE voltage is set to be the same as the MESSAB +ve voltage by means of the recessed screw in the front of the SE card.

Evaluator Adjustments

The only adjustments available on the evaluator are the amplifier gain switch on each LTV-card. Adjustment affects the 5V d.c. voltages, which are the incoming a.c. voltages amplified and rectified. They are measured on the BUPL card test points. DO NOT alter these gain settings without good reason.

Repairs and Renewals

Count heads and their cables must not be separated. If either becomes damaged, the head and cable must be replaced together. DO NOT shorten head cable and re-terminate in an attempt to eliminate even a short piece of damaged cable; the whole cable with head must be replaced. (See Section 2 Page 2)

The following items are interchangeable: EAK cards of the same identification number, evaluator cards of the same identification number.

Axle Counters FLMF
Section 5
Resetting and Restoring the System

RESETTING AND RESTORING THE SYSTEM

This section looks at the processes and procedures for resetting and restoring an axle counter system. It is important to read these notes in conjunction with the most recent instructions issued.

On completion of this section you will be able to:

1. Define the terms reset and restore.

2. Explain the procedure for resetting and restoring an axle counter system.

3. Apply the procedure and complete the restoration form correctly.

DEFINITIONS

Reset	-	To return the electronics to Section Clear, including the counter to zero if not already there.
Restore	-	To reconnect the axle counter to the signalling system, but ONLY WITH THE SIGNALMAN'S PERMISSION.

CIRCUIT OPERATION

The axle counter system has three conditions:

'Section Clear'
'Section Occupied'
'Fault'.

The TPR is de-energised for the conditions 'Section Occupied' and 'Fault'. The system can, without human intervention, switch from:

Clear to Occupied (TPR down)
Occupied to Clear (TPR up)
Clear to Fault (TPR down)
Occupied to Fault (TPR down).

The system CANNOT go from Fault to Occupied or from Fault to Clear on its own.

For the system to be able to show Section Clear after a Fault, it has to be put to Occupied first. Pressing the Reset Plunger energises the TZR. When the plunger is released, the resetting of the electronics is completed, so it shows Clear (Green LED on SIPL card) and the GF relay on the WDH card energises. This, in turn, energises the TSR via TZR UP. TZR drops a short time after the reset plunger is released. TZR DOWN plus TSR UP energises the TPR. (See figure 3.6.)

Restoration procedure

As already stated, the requirements of the current Network Rail “Group Standard” must be observed. These refer to the Reset and Restoration Form.

The following is a Summary of the procedure for filling in the Restoration Form: -

1. Ensure the axle counter is in full working order up to, but excluding, the TPR.
2. Complete Part 1 of the prerequisite proforma (Restoration of axle counter) and assure the Signalman that work is complete.
3. The Signalman will check the state of the Section and, if Clear, will fill in Part 2 and dictate it to you.
4. Record Part 2 as dictated and Restore the axle counter (by reconnecting the TPR coil circuit). Check that the TPR is UP.
5. Record Part 3 of the form and dictate this to the Signalman.
6. Put the form in the appointed place adjacent to the axle counter evaluator.

Axle Counters FLMF
Section 6
Fault Finding and Replacing Equipment

FAULT - FINDING AND REPLACING EQUIPMENT

This section looks at the safety precautions which you must take, the process of fault location and the procedures for replacing the various items of equipment.

On completion of this section you will be able to:

1. Describe some common causes of right-side failures.
2. Take the necessary safety precautions before and during fault-finding.
3. Select and use the test equipment correctly.
4. Locate a fault safely and efficiently.
5. Rectify a fault.

COMMON CAUSES OF RIGHT-SIDE FAILURES

Axle counter systems are extremely reliable, so it is hardly surprising that one of the most common causes of failure is not due to a problem with the axle counter equipment itself, but a power supply failure. This may be the loss of the mains power supply or a failure of the power supply unit, such as the Bauch (or Duvine) unit. When power is reinstated, the system remains in the failed state and has to be reset. (This is to ensure that there are no undetected trains in section.) Other causes include failure of circuit cards in the junction box or the evaluator (both rare), or physical damage to the lineside equipment, particularly the count heads, cables from count heads to junction box, or the line pair. Misalignment of the count heads, perhaps due to a passing train, may also cause a failure. Intermittent failures could indicate a loose count head or termination, or a transmission line fault.

SAFETY PRECAUTIONS AND GENERAL GUIDELINES

The axle counter system must be isolated from the signalling circuitry first. To isolate the system, disconnect the feed to the Section TPR in a suitable manner. The current standard is a 2BA link, wired in series with the TPR coil.

Isolating the axle counter from its TPR means you will be able to use the Technician's Reset Plunger at any time whilst fault-finding, without the risk of the Section showing a false Clear.

Before carrying out any tests, study and record the card indications. It is important to make a note of all evaluator indications before attempting to reset the system, particularly if the failure is of an intermittent nature when some indications may be only transitory.

After that, you may take voltage readings to determine the general nature of the fault, e.g. external fault, card fault, power supply failure.

THE TEST EQUIPMENT

The test equipment was introduced in Section 4. The same meter, dummy wheel and test kit are used in fault-finding. Where more complex faults exist, the SEL Test Set and an oscilloscope may be required. This kind of testing is usually carried out by second-line faulting staff or technical support.

TIPS FOR LOCATING A FAULT

Symptom	Notes
LTV H1 (Red) lit	A count head (MESSAB) voltage is low, i.e. one of the green LEDs on an SE card is out – not due to wheel presence - so monitor frequency not transmitted from lineside. However, H1 will not light if one or both of the two yellow LEDs are out. System may still detect and count even when lit.
LTV H1(Yellow) and H2 (Yellow) - one LED constantly out	These indicate that the a.c. from the line pair for each count head is present and at a useable level. If one is constantly out, this indicates a problem with the corresponding count head. Check the BUPL card voltage, e.g. 1a, 1b, etc. If the BUPL voltage is approx. 1 Volt (background voltage) then go to the detection point and check the equipment there.
LTV H1 (Yellow) and H2 (Yellow) - both constantly out	There is a detection point general fault, e.g. problem with the power supply from evaluator or local source, transmission line fault, card failure, or both count heads defective. Check to see if the d.c. supply is feeding out from the evaluator (where appropriate) on transmission line, also check for composite a.c. signal received from EAK on transmission line. It makes sense to do these tests ***before*** proceeding to the detection points.
DIS G1 (Red) permanently lit	This indicates that the a.c. from the line pair for SK1 (count in) or SK2 (count out) is not being received due to a line disconnection. Reset is not possible with this lit. Check BUPL card 5V (1a, 1b, etc.). Then proceed with same routine given for LTV cards.
DIS G2 (Red) permanently lit	This indicates that the a.c. from the line pair for SK2 (count in) or SK1 (count out) is not being received due to a line disconnection. Reset is not possible with this lit. Check as for G1.
DIS G3 (Red) or DIS G4 (Red) lit - not due to wheel presence being registered	If, on arrival, G3 and/or G4 are lit, but G1 and G2 are out, this could mean a detection point intermittent fault, e.g. loose count head(s) or transmission line problems, usually leading to mis-counting. It could also mean that a lump of metal has influenced a Count Head at some time, WITHOUT a train having passed over the head since.

Symptom	Notes
UMO G12 (Red) lit	This indicates that there has been a power supply interruption to the counter. This indication 'sticks' in the fault state, therefore Reset is necessary. Reset is possible if power has been reinstated and all circuits are functioning correctly.
KTR G24 (Red) or G26 (Red) lit	These indicate a counting fault in counting check store 1 (G24) or 2 (G26). See also DIS card indications.
ZIANZG G2 (Red) out	This indicates that the count supervision was not successful.
ZIANZG G3 (Green) out and track is clear	System is not functioning correctly.
SIPL G2 (Red) lit	This indicates that the track is occupied or that a fault condition exists. Note that, together with G1 (Green) which indicates that the track is clear, this LED monitors the electronics and ***not*** the WDH card relays.

REPLACING EQUIPMENT

Renewal or Re-Termination of Cables

If the fault requires the renewal or re-termination of cables, first remove the relevant SIPL card line fuse.

When the necessary re-cabling or repair has been carried out and the system is ready for final adjustment and testing, the SIPL card fuse should be replaced. The Section TPR must remain disconnected for the time being. Carry out relevant steps in SMTH Test Plan.

It is *important* that, where re-cabling or re-termination has been carried out, a final check is made as follows:

1. Pass the dummy wheel over each count head in turn in the NORMAL direction of travel.
2. Check that the LEDs on SE 01 card are influenced BEFORE those on SE 02 card.
3. If the evaluator is being observed and this test is done at the *count in* end of the Section, check that the evaluator counts UP. If this test is done at the *count out* end, check that the evaluator counts DOWN.
4. With the fault now rectified, check the levels of the BUPL card jack voltages at the evaluator. Adjust as necessary, BUT see Section 4 Pages 4 & 7.

After all work is complete, ensure that the covers on all equipment are in place. Carry out the reset and restoration procedures.

Removal and Replacement of Cards

This section applies during fault-finding ONLY. Below are the suggested procedures for the removal and replacement of cards in the junction box and evaluator.

Removal and replacement of cards in the junction box

1. Loosen the four hexagon-head screws which hold the junction box [EAK] cover in place, then lift the cover upwards, being careful to keep it clear of the apparatus.
2. Before removing the EAK, or any card from it, power MUST be taken off. Therefore, first either disconnect Line Terminal 1 (and temporarily insulate) or remove the local 60 volt supply fuse, depending on how the EAK is powered. Note now that the LEDs on both SE cards are extinguished.
3. Take precautions to prevent Line 1 shorting to Line 2 *even if the supply is obtained locally.*
4. If the EAK is to be removed as one unit, disconnect the line(s), the power supply connections (where provided) and the cables which come from the count head coils. *Be sure* that you can recognise which wires are from which connections (S.M.T.H. refers). The complete unit can be removed after unscrewing the four screws which attach it to the base.
5. If a card is to be removed, note that they are held in place by retaining bars at the top and the bottom. Release both of these by undoing the knurled screws. Each screw is captive in its bar. Take care when removing the cards because they are quite tight in their runners.
6. When replacing a card, ensure that it is squarely in its runners and do not press it fully home until you are sure that the flat pins at top and bottom of the front plate will locate properly. When all cards are fully home in the rack, screw the retaining bars into position USING FINGER PRESSURE ONLY.
7. After bolting down the EAK, reconnect the cables coming from the count heads, the power supply (if appropriate) and the line pair(s). Check that the red LED on each of the SE cards is extinguished.
8. Finally, after carrying out the green LED threshold adjustment and other tests as necessary (see notes in Section 4 on aligning and adjusting the count heads), lower the cover carefully over the apparatus and bolt down.

Removal and replacement of cards in the evaluator

1. First, remove the power from the evaluator rack by removing the supply fuse on the B60 busbar; observe the loss of indications.
2. Press downwards on the plastic clip(s) and remove the card. CAUTION: To prevent damage to any static-sensitive components on the card, do not touch any of the electrical connections. Use Special Handling Facilities wherever possible.
3. Replace the suspected faulty card and restore the 60V supply.
4. It is advisable to test the system after *each* card replacement. Carry out SMTH Test Plan
5. The only card which may be inserted and removed with the power *ON*, is the ZIANZG card. The ZIANZG card does *NOT* have to be inserted for the system to work.
6. When you reinstate the power to the evaluator rack, the UMO card LED will indicate that the power has been removed and then reinstated. You will have to carry out the reset procedure before restoring the system to normal working.